NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR304105
Lead Applicant	Dr Jignesh Patel
Research Title	Making heparin safer - developing and implementing an effective intravenous unfractionated heparin dosing and monitoring protocol into clinical practice
Plain English Summary	**Background** Blood thinning medicines (anticoagulants) are used to treat blood clots. Unfractionated heparin (UFH) is a blood thinning medication. UFH is given to patients by an intravenous (within a vein) drip, particularly in hospitalised patients who are critically unwell, those with kidney disease and patients who undergo heart surgery. UFH offers advantages over other blood thinners - its effects wear off quickly once the intravenous drip is stopped. UFH requires frequent monitoring (up to every 4 hours) to understand if it is working. This is through a blood test. Changes in the UFH dose is guided by the blood test results. In the UK and around the world, use of UFH can cause serious bleeding problems and even death. The way UFH is dosed and monitored are thought to contribute to this. We now know the older method of working out the best UFH dose based on a specific blood test, is not ideal. There is now a more accurate blood test and more up-to-date methods to work out the best UFH dosage. These are known as Population Pharmacokinetic Modelling. In my Fellowship I will use the newer blood test and method to work-out how best to dose and monitor UFH. **Aim** To develop a UFH dosing and monitoring decision chart (protocol) using Population Pharmacokinetic Modelling that can be safely implemented in hospitals, ensuring that patients requiring this high-risk medication will be safer and better carer for. **Method** I will conduct three studies: Study 1 – I will analyse UFH doses and blood test results from patients who received UFH via a drip over a 3-year period at multiple NHS hospitals. Using the Population Pharmacokinetic Modelling method, I will work out the best way to dose and monitor patients with UFH. Study 2 – using the National Reporting and Learning System which records all adverse incidents from hospitals in England and Wales, I will identify and analyse those related to UFH over a 10-year period. This will help to identify and learn from common problems associated with the use of UFH. This information will help develop a new UFH protocol. Study 3 - using the results from studies 1&2, with a team of nurses, doctors, pharmacists, carers, members of the public, patients, porters and laboratory staff, I will co-develop a UFH protocol. I will then implement this in multiple hospitals using the principles of implementation science, a field of science that addresses the adoption of evidence-based good practice into clinical practice. I will then study the success of this implementation. **Patient and Public Involvement (PPI)** Patients and members of the public have inputted into the research planned and confirmed its importance and their views taken on board. Through PPI events, I have established a PPI panel who will co-develop the UFH protocol. This panel will also help co-produce patient facing materials and raise awareness and information on why, when and how UFH is used for patients, the benefits of UFH and the importance of the new protocol to benefit patient outcomes. I will be evaluating how PPI has worked at the end of my fellowship. **Dissemination** I will: -establish a study website where patients, members of the public and health professionals can see how the studies are progressing and what is found and developed and how it is implemented -publish the results in medical journals and present the findings at conferences -work closely with organisations such as Thrombosis UK, the British Society of Haematology, and the Anticoagulation Forum, to ensure widespread dissemination and uptake of the resultant work in the UK and across the world.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

Table Of Contents

1. HEE/NIHR ICA ACAF Introduction
2. Application Summary Information
3. Applicant CV
4. Applicant Research Background
5. Plain English Summary of Research
6. Scientific Abstract
7. Detailed Research Plan
8. Patient & Public Involvement
9. Training & Development and Research Support
10. Detailed Budget
11. Uploads
12. Participants and Signatories
13. Acknowledgement, Review and Submit
14. Validation Summary

1. HEE/NIHR ICA ACAF Introduction

HEE/NIHR ICA Advanced Clinical and Practitioner Academic Fellowship (ACAF) Application Form

Please note that the following information and guidance is intended for candidates submitting an application to the Round 4 (2023) HEE/NIHR ICA ACAF scheme competition ONLY.

Please ensure that you have read the ICA ACAF Round 4 Applicant Guidance Notes (accessible via the left-hand side menu panel) before starting an application and that you refer to them when completing your form - the guidance contains key information about how to complete this online application form. Please note the maximum word limits as indicated throughout the form.

Validating your application
You are advised to validate your application regularly and also between entering information into different sections. You will find a 'Validation Summary' option in the left hand menu. This section will detail any points within your application that are either incomplete or incorrect. Failing to validate your answers may result in you being unable to submit your application by the required deadline.

The Schedule of Events Cost Attribution Tool (SoECAT)
Applications proposing the recruitment of participants may be required to append a completed SoECAT. The SoECAT form needs to be signed off by an AcoRD specialist even where there are no excess treatment costs. There is further guidance for the completion of the form in the Applicant Guidance Notes. The SoECAT form can be uploaded in Excel format. If you are not including a SoECAT form, this must be justified in the application form under the Detailed Research Plan.

The deadline for receipt of applications is 1:00pm on Wednesday 15th November 2023.

Applicant

You will need to complete all of the mandatory sections of the form and enter under the 'Participants and Signatories' section the names and contact details of those individuals fulfilling Participant and Signatory roles (see below).

Once you have completed your sections of the form, and the Participants and Signatories have confirmed their involvement and completed their relevant sections, you will be required to 'Submit' the application to the Signatories for final sign off before the closing date.

You will only be able to 'Submit' your application for final sign off by the signatories when:

- All mandatory sections of the application form are complete.
- All participants and signatories have agreed to be a part of your application.

It is highly recommended that you complete your application well in advance of the deadline to allow the Participants and Signatories enough time to complete their declarations.

Participants and Signatories

You are required to supply the names and email addresses (if not already registered on the ARAMIS application system) of the individuals who will be undertaking Participant and Signatory roles as part of your application. Participants and Signatories must agree to be part of the application before it is submitted.

Participants and Signatories are required to review the declaration for their role before confirming participation. By confirming participation, Participants and Signatories are acknowledging their involvement and input into this application and agree to be involved in it. You must ensure that all Participants and Signatories are happy for your application to be submitted before submitting it on the online system.

Required Participants:

Research Support: Applicants are required to enter details of at least 1 (and up to 3) individuals who will provide research support (mentorship) during the Fellowship. The individual(s) acting as the applicant's Research Support/ Mentor must confirm that they have read the application and the Applicant Guidance Notes and are willing to act

as the applicant's mentor for research and career development and agree to abide by the conditions under which an award may be granted.

Practice Support: Applicants are required to enter details of at least 1 (and up to 3) individuals who will provide professional practice support/ mentorship during the Fellowship. The individual(s) acting as the applicant's Practice Support/ Mentor(s) must confirm that they have read the application and the Applicant Guidance Notes and are willing to provide professional and career development support to the applicant and agree to abide by the conditions under which an award may be granted.

Host Organisation Administrative Authority or Finance Officer: The Administrative Authority or Finance Officer for the Host Organisation must confirm that they will ensure the accuracy of the financial details of the application and that the Host Organisation is prepared to carry out this research at the stated costs and to administer the award if made.

Required Signatories:

Host Organisations' Heads of Department: Applicants are required to enter the details of the relevant Head of Departments at both the Higher Education Institute (HEI) and the provider of health or care services in which this award will be based (the Host Organisation and the Partner Organisation). In agreeing to participate in this application, the Heads of Department are confirming that they support the application and that, if funded, the research and training will be supported and administrated in the named organisations and that the applicant for whom they are responsible will undertake the work.

The Heads of Department are required to provide a joint statement of support via the 'Training & Development and Research Support' section of the application form, which will be accessible to them once they have confirmed participation.

Application Submission

Once the application is ready (see list of required steps under the 'applicant' heading above), applicants are able to submit the application for final signatory approval. Upon submission by the applicant, the individuals added in the Signatory roles will be prompted to sign back into the system and approve the finalised application.

The application will not be complete until all the required signatories have approved the final version of the form submitted by the applicant. Only when the last signatory presses the 'Approve' button will the completed application be submitted to the NIHR.

Please note, all of the steps described here, including signatory approval of the submitted application, must take place before the deadline of **1:00pm on Wednesday 15th November 2023.** No extensions will be given and no exceptions made to allow for completion of the form.

Should you require assistance in completing the online form, please contact the NIHR on 0113 532 8444 or by emailing academy-awards@nihr.ac.uk.

Click on the link below for further information regarding the Application Form Submission Process:

2. Application Summary Information

Host organisation (which will administer any award)
King's College Hospital NHS Foundation Trust

Partner organisation	King's College London

Application Type Please select the relevant Partnership Fellowship.
N/A

Advanced Fellowship application type	Establishing or recently established independence

Research Title	Making heparin safer - developing and implementing an effective intravenous unfractionated heparin dosing and monitoring protocol into clinical practice

Research Type	Other (please specify)

Please provide more details on your selection above.
My fellowship involves 3 research studies: - a population pharmacokinetic multi-centre modelling study of intravenous unfractionated heparin, using both retrospective and prospective data from patients who have/are receiving an intravenous heparin infusion. - a retrospective analysis of all unfractionated heparin incidents reported to the National Reporting and Learning System over a 10 year period. - a pre- and post-implementation study of implementation and evaluation of a contemporary intravenous unfractionated heparin dosing and monitoring protocol into clinical practice, using mixed methodologies.

Proposed start date if grant awarded:	01/10/2024

Do you wish to hold an ACAF for the full time equivalent of 2, 3, 4 or 5 years?
3

WTE of Award	1.00

Duration (months) This is automatically populated based on the number of years and the WTE % selected for the Fellowship
36

Professional Practice Development Please confirm the % of award time you propose to undertake Professional Practice Development

20%

3. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Grant Application System or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

The 'Publication Record' section of the form is automatically populated from publications added into the "My Research Outputs" page of your ARAMIS account.

Professional Bodies

Please ensure that you have provided details of your current registration to enable confirmation of eligibility. If you do not currently hold registration with the body listed in the applicant guidance notes for your stated profession, please provide further information in the "Research Career to Date" section of the application form.

Membership Reference	Institution
2047229	General Pharmaceutical Council

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2013	07/2014	Postgraduate Certificate	Independent Prescribing	United Kingdom	King's College London	Distinction
08/2009	08/2012	PhD	Pharmacokinetics	United Kingdom	King's College London	-
09/1999	06/2001	MSc	Clinical Pharmacy	United Kingdom	University of Brighton	Merit
09/1997	06/1998	Postgraduate Certificate	Clinical Pharmacy	United Kingdom	University College London	Good pass
09/1993	06/1996	BSc(Hons)	Pharmacy	United Kingdom	University of Manchester	First Class

Present and previous positions

(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
09/2018	-	Reader and Honorary Consultant Pharmacist in Anticoagulation	King's College London & King's College Hospital Foundation NHS Trust	33
08/2012	08/2018	Clinical Senior Lecturer and Honorary Consultant Pharmacist in Anticoagulation	King's College London & King's College Hospital Foundation NHS Trust	33
08/2009	07/2012	GSTT/KCL NIHR BRC Allied Healthcare Professional Research Training Fellow	King's College Hospital Foundation NHS Trust	100
01/2008	06/2009	Anticoagulation Pharmacist and Programme Director PG Dip General Pharmacy Practice	King's College Hospital Foundation NHS Trust & UCL School of Pharmacy	0
08/2007	12/2007	Lead Pharmacist (Surgery and Antibiotics) and Programme Director PG Dip General Pharmacy Practice	Mayday Healthcare NHS Trust & UCL School of Pharmacy	0
05/2004	07/2007	Lead Pharmacist (Surgery and Antibiotics) and Teacher-Practitioner	Mayday Healthcare & University of Brighton	0
10/2001	04/2004	Teacher-Practitioner	Brighton and Sussex	0

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
		(Anticoagulation)	University Hospitals NHS Trust & University of Brighton	
05/2000	09/2001	Senior Pharmacist, Medicines Information	Kingston Hospital	0
01/1999	04/2000	Medical Unit Pharmacist	Kingston Hospital	0
07/1997	07/1998	Resident Pharmacist	University Hospital Lewisham	0
07/1996	07/1997	Pre-registration Pharmacist	University Hospital Lewisham	0

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Population Pharmacokinetic – Pharmacodynamic modelling of enoxaparin in pregnancy	Personal research award - funded by the GSTT/KCL NIHR BRC. Chief investigator. Designed the study, collected the data and completed the primary data analysis. Lead author for all publications that resulted from the work. Main output: doi:10.1161/CIRCULATIONAHA.113.003198	232256	GSTT	36	01/08/2009	31/07/2012
The Switching Study	Secured finding. Co-designed the protocol and oversaw the successful completion of the work. Chief investigator for	307800	Bayer plc	48	01/09/2014	31/08/2018

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	the programme of work. Main output: doi: 10.1016/j.thromres.2017.12.021.					
When to thrombolyse patients prescribed concurrent novel oral anticoagulants in acute stroke	Conceived the study and secured funding. Oversaw the completion of the work. Chief investigator. Main output: doi.org/10.1111/ijs.12634_21	7425	King's Health Partners Challenge Fund	12	01/01/2015	31/12/2015
Follow-up In Rivaroxaban patients in Setting of Thromboembolism (FIRST)	Contributed to study design, supervised the work to successful completion. Co-applicant - clinical pharmacology expert on the grant. Main output: doi: 10.1002/rth2.12607	520000	Bayer plc	60	01/09/2014	31/08/2019
The role of apixaban, aspirin and enoxaparin as thromboprophylaxis in patients newly diagnosed with multiple myeloma (TiMM)	Conceived study. Led the grant writing that secured the funding. Supervised the completion of the work. Involved in data collection (focus groups). Main output: doi: 10.1007/s11239-109-01891-	223156	NIHR (RfPB)	24	01/09/2015	31/08/2017

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	0.					
What dose of vitamin D is required to achieve target serum 25-hydroxyvitamin D levels in children?	Primary supervisor to doctoral student: Health Education England/National Institute for Health Research (NIHR) ICA fellowship. Contributed to study design and successfully supervised work to completion. Main output: doi: org/10.1111/bcp.15064	322705	HEE/NIHR	36	01/09/2017	31/08/2020
Rivaroxaban versus warfarin for stroke patients with antiphospholipid syndrome, with or without SLE (RISAPS)	Contributed to the study design relating to the population PK modelling aspects. Collaborator - responsible for the PPK modelling of rivaroxaban in this trial. Trial has recently completed. Awaiting final data to model.	970683	Versus Arthritis	60	01/09/2018	31/08/2023
Population pharmacokinetics of edoxaban	Primary supervisor to PCAF student. Contributed to the study design. Supervised work to successful completion.	69133	HEE/NIHR ICA PCAF	12	01/09/2022	31/08/2023

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
	Paper submitted - under review.					
Developing a medicines optimisation tool for patients with chronic kidney disease	Primary supervisor to the PCAF student. Contributed to study design.	75924	HEE/NIHR ICA PCAF	12	01/07/2023	30/06/2024

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Speed, V., Clapham, R., Bartoli-Abdou, J., Stirling, K., Akor, F., Collings, V., Patel, R., Byrne, R., Arya, R. and **Patel, JP.**. (2023) ‘Factor Xa Inhibitor Plasma Concentrations and Clinical Outcomes in Patients Weighing ≤50 kg-Experience from Four UK Centers.’, *Thrombosis and haemostasis*.
Patel, JP., Thavarajah, V. and Roberts, LN.. (2023) ‘Response to "Clinically relevant increases in the international normalized ratio and model of end-stage liver disease score by therapeutic doses of direct oral anticoagulants in patients with cirrhosis".’, *Research and practice in thrombosis and haemostasis*, 7(4).
Sheikh, U., Ding, A., Nzelu, O., Roberts, LN., Johns, J., Ross, J., Arya, R. and **Patel, JP.**. (2023) ‘The "lived" experience of menstruating women commencing anticoagulants.’, *Research and practice in thrombosis and haemostasis*, 7(3).
Patel, JP., Nzelu, O., Roberts, LN., Johns, J., Ross, J. and Arya, R.. (2023) ‘How do anticoagulants impact menstrual bleeding and quality of life? - The PERIOD study.’, *Research and practice in thrombosis and haemostasis*, 7(2).
Müller, S., Tittl, L., Speed, V., Roberts, L., **Patel, J.**, Patel, R., Arya, R., Kucher, N., Spirk, D., Sahin, K. and Beyer-Westendorf, J.. (2022) ‘Pooled Analysis of Rivaroxaban therapy for acute venous thromboembolism in FIRST registry, SWIVTER and DRESDEN NOAC registry.’, *Research and practice in thrombosis and haemostasis*, 6(7).
Woolcombe, SA., Ball, RE. and **Patel, JP.**. (2022) ‘Managing direct oral anticoagulants in accordance with the Scottish Dental Clinical Effectiveness Programme guidance for patients undergoing dentoalveolar surgery.’, *British dental journal*, 232(8).
Wan, M., **Patel, J.**, Rait, G. and Shroff, R.. (2022) ‘Hypervitaminosis D and nephrocalcinosis: too much of a good thing?’, *Pediatric nephrology (Berlin, Germany)*, 37(10).
Speed, V., Auyeung, V., **Patel, JP.**, Cooper, D., Miller, S., Roberts, LN., Patel, RK. and Arya, R.. (2021) ‘Adherence to rivaroxaban for the treatment of venous thromboembolism-Results from the FIRST registry.’, *Research and practice in thrombosis and haemostasis*, 5(8).
Bartoli-Abdou, JK., **Patel, JP.**, Vadher, B., Brown, A., Roberts, LN., Patel, RK., Arya, R. and Auyeung, V.. (2021) ‘Long-term adherence to direct acting oral anticoagulants and the influence of health beliefs after switching from vitamin-K antagonists: Findings from the Switching Study.’, *Thrombosis Research*, 208.
Speed, V., **Patel, JP.**, Cooper, D., Miller, S., Roberts, LN., Patel, RK. and Arya, R.. (2021) ‘Rivaroxaban in acute venous thromboembolism: UK prescribing experience.’, *Research and practice in thrombosis and haemostasis*, 5(7).
Speed, V., **Patel, J.** and Arya, R.. (2021) ‘Bleeding issues in women prescribed anticoagulation’, *Thrombosis Update*, 5.
Wan, M., Green, B., Iyengar, AA., Kamath, N., Reddy, HV., Sharma, J., Singhal, J., Uthup, S., Ekambaram, S.,

Publication Record

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Publications
Selvam, S., Rait, G., Shroff, R. and **Patel, JP.**. (2021) ‘Population pharmacokinetics and dose optimisation of colecalciferol in paediatric patients with chronic kidney disease.’, *British journal of clinical pharmacology*.
Speed, V., Roberts, LN., Arya, R. and **Patel, JP.**. (2021) ‘Fixed dose rivaroxaban can be used in extremes of bodyweight: A population pharmacokinetic analysis - Reply to Jacobs & Ryan comment.’, *Journal of thrombosis and haemostasis : JTH*, 19(3).
McFarlane, SQ., **Patel, JP.**, Auyeung, V. and Arya, R.. (2021) ‘An international survey of low molecular weight heparin prescribing in the context of antenatal venous thromboembolism.’, *Thrombosis Research*, 200.
Rajaratnam, N., **Patel, JP.**, Roberts, LN., Czuprynska, J., Patel, RK. and Arya, R.. (2021) ‘More on enoxaparin thromboprophylaxis in pregnancy: A review of 10 years' experience from King's College Hospital.’, *Journal of thrombosis and haemostasis : JTH*, 19(1).
Ding, A., **Patel, JP.** and Auyeung, V.. (2020) ‘Testing the Traditional Chinese Medicine Consultation Model for Adherence in Complementary and Alternative Medicine.’, *Evidence-based complementary and alternative medicine : eCAM*, 2020.
Patel, R., Czuprynska, J., Roberts, LN., Vadher, B., Rea, C., Patel, R., Gee, E., Saul, G., Coles, E., Brown, A., Byrne, R., Speed, V., **Patel, JP.** and Arya, R.. (2020) ‘Switching warfarin patients to a direct oral anticoagulant during the Coronavirus Disease-19 pandemic.’, *Thrombosis Research*, 197.
Speed, V., **Patel, JP.**, Roberts, LN., Patel, RK. and Arya, R.. (2020) ‘UK real-world experience of prescribing rivaroxaban for acute venous thromboembolism in obesity: Results from the FIRST Registry.’, *Thrombosis Research*, 197.
Patel, J., Auyeung, V., Cameron, L., Chanda, R., Husain, N., Jubraj, B., Shah, K., Shah, R., Sherikhan, N., Stevenson, JM., Waghorn, J. and Davies, JG.. (2020) ‘Final year M.Pharm. student views and performance in objective structured clinical examinations’, *Pharmacy Education*, 20(1).
Zhao, Y., Arya, R., Couchman, L. and **Patel, JP.**. (2020) ‘Are apixaban and rivaroxaban distributed into human breast milk to clinically relevant concentrations?’, *Blood*, 136(15).
Speed, V., Green, B., Roberts, LN., Woolcombe, S., Bartoli-Abdou, J., Barsam, S., Byrne, R., Gee, E., Czuprynska, J., Brown, A., Duffy, S., Vadher, B., Patel, R., Scott, V., Gazes, A., Patel, RK., Arya, R. and **Patel, JP.**. (2020) ‘Fixed dose rivaroxaban can be used in extremes of bodyweight: A population pharmacokinetic analysis.’, *Journal of thrombosis and haemostasis : JTH*, 18(9).
Wan, M., Patel, A., **Patel, JP.**, Rait, G., Jones, SA. and Shroff, R.. (2020) ‘Quality and use of unlicensed vitamin D preparations in primary care in England: Retrospective review of national prescription data and laboratory analysis.’, *British journal of clinical pharmacology*, 87(3).
Wan, M., Patel, A., **Patel, JP.**, Rait, G., Jones, SA. and Shroff, R.. (2020) ‘Quality and use of unlicensed vitamin D preparations in primary care in England: Retrospective review of national prescription data and laboratory analysis.’, *British journal of clinical pharmacology*, 87(3).
Zhao, Y., Couchman, L., Kipper, K., Arya, R. and **Patel, JP.**. (2020) ‘A UHPLC-MS/MS method to simultaneously quantify apixaban, edoxaban and rivaroxaban in human plasma and breast milk: For emerging lactation studies.’, *Journal of chromatography. B, Analytical technologies in the biomedical and life sciences*, 1144.
Ding, A., **Patel, JP.** and Auyeung, V.. (2020) ‘Understanding the Traditional Chinese Medicine (TCM) consultation: Why do patients adhere to treatment?’, *Complementary therapies in clinical practice*, 39.
Guimicheva, B., Roberts, LN., **Patel, JP.**, Subramanian, D. and Arya, R.. (2020) ‘Mode of Delivery Does Not Influence Postpartum Hypercoagulability Measured by Thrombin Generation or Thromboelastometry.’, *TH open : companion journal to thrombosis and haemostasis*, 4(1).
Wan, M., Horsfall, LJ., Basatemur, E., **Patel, JP.**, Shroff, R. and Rait, G.. (2019) ‘Vitamin D prescribing in children in UK primary care practices: a population-based cohort study.’, *BMJ open*, 9(12).
Sayar, Z., Czuprynska, J., **Patel, JP.**, Benjamin, R., Roberts, LN., Patel, RK., Cornelius, V. and Arya, R.. (2019) ‘What are the difficulties in conducting randomised controlled trials of thromboprophylaxis in myeloma patients and how can we address these? Lessons from apixaban versus LMWH or aspirin as thromboprophylaxis in newly

Publication Record

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Publications
diagnosed multiple myeloma (TiMM) feasibility clinical trial.’, *Journal of thrombosis and thrombolysis*, 48(2).
Mabley, J., **Patel, JP.**, Sayed, A., Arya, R. and Scutt, G.. (2019) ‘Direct oral anticoagulant (DOAC)-mediated vasodilation: Role of nitric oxide.’, *Thrombosis Research*, 176.
Patel, JP., Byrne, RA., Patel, RK. and Arya, R.. (2019) ‘Progress in the monitoring of direct oral anticoagulant therapy.’, *British journal of haematology*, 184(6).
Guimicheva, B., **Patel, JP.**, Roberts, LN., Subramanian, D. and Arya, R.. (2018) ‘Women's views, adherence and experience with postnatal thromboprophylaxis.’, *Thrombosis Research*, 173.
Speed, V., Roberts, LN., **Patel, JP.** and Arya, R.. (2018) ‘Venous thromboembolism and women's health.’, *British journal of haematology*, 183(3).
Zhao, Y., Ding, A., Arya, R. and **Patel, JP.**. (2018) ‘Factors influencing the recruitment of lactating women in a clinical trial involving direct oral anticoagulants: a qualitative study.’, *International Journal of Clinical Pharmacy*, 40(6).
Byrne, R., Brown, A., **Patel, JP.**, Czuprynska, J., Roberts, LN., Patel, RK. and Arya, R.. (2018) ‘Sub therapeutic rivaroxaban plasma concentrations following administration via Percutaneous Endoscopic Gastrostomy (PEG) feeding tubes - A note of caution.’, *Thrombosis Research*, 168.
Sayar, Z., Speed, V., **Patel, JP.**, Patel, RK. and Arya, R.. (2018) ‘The perils of inhibiting deficient factors. ’, *Journal of thrombosis and haemostasis : JTH*.
Bartoli-Abdou, JK., **Patel, JP.**, Xie, R., Dzahini, O., Vadher, B., Brown, A., Roberts, LN., Patel, RK., Arya, R. and Auyeung, V.. (2018) ‘Associations between illness beliefs, medication beliefs, anticoagulation-related quality of life, and INR control: Insights from the Switching Study.’, *Research and practice in thrombosis and haemostasis*, 2(3).
Patel, JP., Chitongo, PB., Dighe, P., Roberts, LN., Vadher, B., Patel, RK. and Arya, R.. (2018) ‘Prothrombin times in the presence of edoxaban - in-vivo experience from King's College hospital.’, *British journal of haematology*, 184(3).
Bartoli-Abdou, JK., **Patel, JP.**, Crawshaw, J., Vadher, B., Brown, A., Roberts, LN., Patel, RK., Arya, R. and Auyeung, V.. (2018) ‘Exploration of adherence and patient experiences with DOACs one year after switching from vitamin-K antagonists- insights from the switching study.’, *Thrombosis Research*, 162.
Ding, A., **Patel, J.** and Auyeung, V.. (2018) ‘Attitudes and Beliefs that Affect Adherence to Provider-Based Complementary and Alternative Medicine: A Systematic Review’, *European Journal of Integrative Medicine*, 17.
Patel, J., Roberts, L., Patel, R. and Arya, R.. (2018) ‘Pregnancy outcomes in women with mechanical prosthetic heart valves - a prospective descriptive population based study using the United Kingdom Obstetric Surveillance System (UKCOSS) data collection system - comment on use and monitoring of low molecular weight heparin in this setting.’, *British Journal of Obstetrics and Gynaecology*, 125(1).
Barsam, SJ., **Patel, JP.**, Roberts, LN., Kavarthapu, V., Patel, RK., Green, B. and Arya, R.. (2017) ‘The impact of body weight on rivaroxaban pharmacokinetics.’, *Research and practice in thrombosis and haemostasis*, 1(2).
Patel, JP., Roberts, LN., Patel, RK. and Arya, R.. (2017) ‘Re: Pregnancy outcomes in women with mechanical prosthetic heart valves - a prospective descriptive population-based study using the United Kingdom Obstetric Surveillance System (UKOSS) data collection system.’, *BJOG : an international journal of obstetrics and gynaecology*, 125(1).
Czuprynska, J., **Patel, JP.** and Arya, R.. (2017) ‘Current challenges and future prospects in oral anticoagulant therapy.’, *British journal of haematology*, 178(6).
Omunakwe, HE., Roberts, LN., **Patel, JP.**, Subramanian, D. and Arya, R.. (2017) ‘Re: A comparison of the recommendations for pharmacologic thromboembolism prophylaxis after caesarean delivery from the major guidelines: Impact on thromboprophylaxis rates of implementing Royal College of Obstetricians and Gynaecologists' guidance for reducing the risk of ante- and postnatal venous thromboembolism.’, *BJOG : an international journal of obstetrics and gynaecology*, 124(5).
Patel, JP., Woolcombe, SA., Patel, RK., Obisesan, O., Roberts, LN., Bryant, C. and Arya, R.. (2017) ‘Managing direct oral anticoagulants in patients undergoing dentoalveolar surgery.’, *British dental journal*, 222(4).

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Lock, J., Patel, RK., Brookes, C. and **Patel, JP.**. (2016) ‘Clinical Experience of Prescribing Dabigatran Etexilate With Dronedarone: The Role of Therapeutic Drug Monitoring.’, *Therapeutic drug monitoring*, 38(6).
Auyeung, V., **Patel, JP.**, Abdou, JK., Vadher, B., Bonner, L., Brown, A., Roberts, LN., Patel, RK. and Arya, R.. (2016) ‘Anticoagulated patient's perception of their illness, their beliefs about the anticoagulant therapy prescribed and the relationship with adherence: impact of novel oral anticoagulant therapy - study protocol for The Switching Study: a prospective cohort study.’, *BMC hematology*, 16(1).
Abdou, JK., Auyeung, V., **Patel, JP.** and Arya, R.. (2016) ‘Adherence to long-term anticoagulation treatment, what is known and what the future might hold.’, *British journal of haematology*, 174(1).
Watras, MM., **Patel, JP.** and Arya, R.. (2016) ‘Traditional Anticoagulants and Hair Loss: A Role for Direct Oral Anticoagulants? A Review of the Literature.’, *Drugs - real world outcomes*, 3(1).
Patel, J.. (2015) ‘Monitoring of non-vitamin K antagonist oral anticoagulants: is there a clinical need?’, *Thrombus*, 19(4).
Ferreira, M., Barsam, S., **Patel, JP.**, Czuprynska, J., Roberts, LN., Patel, RK. and Arya, R.. (2015) ‘Heavy menstrual bleeding on rivaroxaban.’, *British journal of haematology*, 173(2).
Sharma, M., Cornelius, VR., **Patel, JP.**, Davies, JG. and Molokhia, M.. (2015) ‘Efficacy and Harms of Direct Oral Anticoagulants in the Elderly for Stroke Prevention in Atrial Fibrillation and Secondary Prevention of Venous Thromboembolism: Systematic Review and Meta-Analysis.’, *Circulation*, 132(3).
Patel, JP., Couchman, L., Chitongo, PB., Flanagan, RJ. and Arya, R.. (2015) ‘New oral anticoagulants: dosing and monitoring.’, *BMJ (Clinical research ed.)*, 350.
Bouchard, J., Ghannoum, M., Bernier-Jean, A., Williamson, D., Kershaw, G., Weatherburn, C., Eris, JM., Tran, H., **Patel, JP.** and Roberts, DM.. (2015) ‘Comparison of intermittent and continuous extracorporeal treatments for the enhanced elimination of dabigatran.’, *Clinical toxicology (Philadelphia, Pa.)*, 53(3).
Patel, JP., Patel, RK., Roberts, LN., Marsh, MS., Green, B., Davies, JG. and Arya, R.. (2014) ‘Changes in thrombin generation and D-dimer concentrations in women injecting enoxaparin during pregnancy and the puerperium.’, *BMC pregnancy and childbirth*, 14.
Patel, JP., Chitongo, PB., Czuprynska, J., Roberts, LN., Patel, RK. and Arya, R.. (2014) ‘Normal prothrombin times in the presence of therapeutic levels of apixaban--in-vivo experience from King's College Hospital.’, *British journal of haematology*, 169(1).
Gous, T., Couchman, L., **Patel, JP.**, Paradzai, C., Arya, R. and Flanagan, RJ.. (2014) ‘Measurement of the direct oral anticoagulants apixaban, dabigatran, edoxaban, and rivaroxaban in human plasma using turbulent flow liquid chromatography with high-resolution mass spectrometry.’, *Therapeutic drug monitoring*, 36(5).
Patel, JP. and Arya, R.. (2013) ‘The current status of bridging anticoagulation.’, *British journal of haematology*, 164(5).
Patel, JP., Green, B., Patel, RK., Marsh, MS., Davies, JG. and Arya, R.. (2013) ‘Population pharmacokinetics of enoxaparin during the antenatal period.’, *Circulation*, 128(13).
Patel, JP., Roberts, LN., Chitongo, PB., Patel, RK. and Arya, R.. (2013) ‘More on normal prothrombin times in the presence of therapeutic levels of rivaroxaban--early experience from King's College Hospital.’, *British journal of haematology*, 162(5).
Patel, JP., Auyeung, V., Patel, RK., Marsh, MS., Green, B., Arya, R. and Davies, JG.. (2012) ‘Women's views on and adherence to low-molecular-weight heparin therapy during pregnancy and the puerperium.’, *Journal of thrombosis and haemostasis : JTH*, 10(12).
Patel, JP., Green, B., Patel, RK., Davies, JG. and Arya, R.. (2012) ‘Population pharmacokinetic analysis of the oral thrombin inhibitor dabigatran etexilate in patients with non-valvular atrial fibrillation from the RE-LY trial: a rebuttal.’, *Journal of thrombosis and haemostasis : JTH*, 10(3).
Patel, JP., Roberts, LN. and Arya, R.. (2011) ‘Anticoagulating obese patients in the modern era.’, *British journal of haematology*, 155(2).

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Publications
Patel, JP., Patel, RK., Davies, JG. and Arya, R.. (2011) ‘Prophylaxis with low-dose low molecular weight heparin during pregnancy and the puerperium: is it effective? A rebuttal.’, *Journal of thrombosis and haemostasis : JTH*, 9(6).
Patel, JP., Sharma, A., West, D., Bates, IP., Davies, JG. and Abdel-Tawab, R.. (2011) ‘An evaluation of using multi-source feedback (MSF) among junior hospital pharmacists.’, *The International journal of pharmacy practice*, 19(4).
Patel, J.. (2011) ‘Prophylactic dosing of Low Molecular Weight Heparins (LMWH) in obese patients’, *Thrombus*, 15(2).
James, KL., Davies, JG., Kinchin, I., **Patel, JP.** and Whittlesea, C.. (2010) ‘Understanding vs. competency: the case of accuracy checking dispensed medicines in pharmacy.’, *Advances in health sciences education : theory and practice*, 15(5).
Patel, J.. (2010) ‘Give juniors mini PATs on the back’, *Clinical Pharmacist*, 2.
Patel, JP., West, D., Bates, IP., Eggleton, AG. and Davies, G.. (2009) ‘Early experiences of the mini-PAT (Peer Assessment Tool) amongst hospital pharmacists in south east London.’, *The International journal of pharmacy practice*, 17(2).
Patel, JP. and Hunt, BJ.. (2008) ‘Where do we go now with low molecular weight heparin use in obstetric care?’, *Journal of thrombosis and haemostasis : JTH*, 6(9).
Gilchrist, M., Franklin, BD. and **Patel, JP.**. (2008) ‘An outpatient parenteral antibiotic therapy (OPAT) map to identify risks associated with an OPAT service.’, *The Journal of antimicrobial chemotherapy*, 62(1).
Corbo, **Patel, J.**, Abdel-Tawab, R. and Davies, JG.. (2006) ‘Evaluating clinical skills of undergraduate pharmacy students using objective structured clinical examinations (OSCEs)’, *Pharmacy Education*, 6(1).
Ramsay, NA., Kenny, MW., Davies, G. and **Patel, JP.**. (2005) ‘Complimentary and alternative medicine use among patients starting warfarin.’, *British journal of haematology*, 130(5).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/07/2009	GSTT/KCL NIHR BRC	PhD research training fellowship	Chief investigator
01/04/2009	United Kingdom Clinical Pharmacy Association (UKCPA)	Winner of the UKCPA Education and Training Award: - Is there a place for the mini Peer Assessment Tool (PAT) amongst general level pharmacists working in secondary care?	Chief investigator

ORCID iD 0000-0003-4197-8294

4. Applicant Research Background

Professional Background	Other Registered Health or Care Professional

Profession (Other Registered Health or Care Professional)
Pharmacist

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
2047229 - GPhC

Please describe your research career to date

Research experience

My PhD defined the optimal dosing strategy of enoxaparin for the management of venous thromboembolism (VTE) during pregnancy, using population pharmacokinetic modelling (PPM) methodology. The work led to a change in the Royal College of Obstetricians and Gynaecologists clinical guidelines for the anticoagulation management of antenatal VTE. Through this work, I learnt and applied the method of non-linear mixed effects pharmacokinetic modelling and developed specific skills in the use R, NONMEM and SPSS.

Since my PhD, I have led several research studies as chief investigator (CI) and principal investigator (PI), including clinical trials. My research programme focuses on the safe and effective use of anticoagulation therapy. Examples of studies I have led include (i) PPM of rivaroxaban (CI), (ii) The Switching study (CI), (iii) Apixaban and rivaroxaban in breastfeeding (PI), (iv) The PERIOD study (CI) and (v) Thromboprophylaxis in Multiple Myeloma (NIHR PB-PG-0214-33101) (PI).

Research impact & current work

My research has optimised the use of anticoagulants in under-served patient populations, has identified and minimised new risks, and addressed uncertainties around the use of both traditional and newer classes of anticoagulant in patients including the elderly, menstruating women, obese patients, breastfeeding women, and those undergoing oral surgery. My work has raised awareness of risks of anticoagulants amongst clinicians, changed clinical practice guidelines, and has led to safer and more effective prescribing, ultimately improving patient safety and quality of life. For example, the recently completed PERIOD study, where I have been invited to speak on the findings and implications at British Society of Haematology (BSH) (Apr 2023), VTE Dublin (Oct 2023) and at the International Society of Thrombosis and Haemostasis conference in Thailand (June 2024).

The translational nature of my research has consistently been returned in REF rounds, with my research specifically featuring in King's College London's REF 2021 UoA3 submission, which received an outstanding evaluation: 1st in the country for research quality and power. My research was specifically highlighted in the UoA3 Environment Statement.

I am currently leading and am PI on the New Horizons Clinical Trial (https://www.isrctn.com/ISRCTN57583139) – a phase IV clinical trial, evaluating the transfer of edoxaban and rivaroxaban into human breastmilk.

Since July 2021, I have been the Medicines Use research group lead, in the School of Cancer and Pharmaceutical Sciences, at KCL. I have published >70 peer-reviewed publications and 2 book chapters.

Due to my expertise in anticoagulation, I have been co-opted to the BSH guideline writing group of: Measurement of Heparin and other non-Coumarin anticoagulants and their effects on tests of Haemostasis.

Research capacity building

As a clinical academic, growing research capacity within the health service has been a priority for me. I have successfully supervised 7 doctoral students (5 PhD and 2 MD(Res)) to completion, 5 as primary supervisor. I am currently supervising 3 further students, a consultant pharmacist (PhD), a ST5 in gynaecology (MD(Res)) and a ST5 in haematology (PhD). I was primary supervisor for a NIHR doctoral student, a pharmacist (ICACDRF-2016-02-057) through their PhD award and supported them through their recent skills / development award (NIHR302279). Another pharmacist (NIHR302672) has just completed a PCAF with me and has now submitted a DCAF application. A further pharmacist (NIHR303436) has just started her PCAF with me from September 2023. I am the pharmacy representative on the Health Education England – London, Integrated Clinical Academic Pre-Doctoral Bridging Programme, run by KCL and London South Bank University. I am a member of the Department of Health, Chief Pharmaceutical Officer's short life working group of Clinical Academic Career Pathways in Pharmacy.

I was commissioned by the Royal Pharmaceutical Society of Great Britain, funded by the NIHR, to develop a bespoke e-learning module for developing research capability and capacity amongst pharmacists and pharmacy technicians. I led the development of the 'Reporting, publishing, dissemination, and impact' module which launched in 2023.

I am a member of the Scientific Advisory Panel of Pharmacy Research UK, a charity which funds capacity building and original research to help pharmacists and pharmacy improve the public's health and deliver better outcomes for patients.

Research delivery

Between April 2014 and April 2017, I was the Non – Malignant Haematology Clinical Lead for the NIHR South London Clinical Research Network. My role involved working with local PIs to ensure recruitment targets were being met, engaging with clinicians who were not research active to facilitate research within their organisations, and representing South London haematology nationally. During my tenure, Non-Malignant Haematology in South London exceeded recruitment targets set and I was the only non-medical lead in the National group, and the only pharmacist to date to be appointed to such a position in any speciality.

I have reviewed grants for the: NIHR, Stroke Association, Wellcome Trust, and referred papers for: Lancet Haematology, Thrombosis Research, Clinical Pharmacokinetics, D&T Bulletin, British Journal of Haematology, The Cochrane Review Collaborative, Clinical Pharmacology&Therapeutics, PLOS Medicine, British Dental Journal.

Reason for fellowship application

I am a national leader relating to research and practice in the safe and effective use of anticoagulants. This fellowship and the work outlined in my fellowship application will afford me the time and space to tackle the problem of dosing, monitoring, and safety of unfractionated heparin, the results of which will directly translate to clinical practice in the NHS. The fellowship will also provide me with specific training and experience in the field of implementation science; an area which I have little knowledge and skill in, but which holds huge potential in ensuring the translation of the work I undertake both through this fellowship and in the future.

Impact on career

I aim to be an international research leader in the field of anticoagulation therapy research and practice. This fellowship and the training received will provide me with the required platform to achieve this and develop and refine my personal skills and attributes required to function effectively at this level.

Title of research degree	PhD in Pharmacokinetics

Has the degree already been awarded?	Yes

Date of Award	01/03/2013

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
I applied through an NIHR Advanced Fellowship (round 9) for this project - ref: NIHR303192. Key feedback received on not progressing the application by the reviewers was focussed on the single centre nature of the original application - whereby the original application was developing the pharmacokinetic model and implementing the UFH protocol developed in a single centre study, at King's College Hospital only. The reviewers also highlighted the implementation aspects of the work required more details. Taking this feedback on-board, the proposed research is now multi-centre, with Guy's and St. Thomas' Hospital and the Royal Brompton and Harefield Hospitals. This means that in total 6 hospital sites which will be involved in my fellowship work (KCH - Denmark Hill, KCH - Princess Royal Hospital, Guy's Hospital, St.Thomas' Hospital, Royal Brompton Hospital and Harefield Hospital). These partners will enhance the work proposed and ensure the credibility of the final heparin dosing and monitoring protocol developed. All these hospital sites will be implementing the UFH protocol. Details on the pre and post-implementation studies are now fully provided, with the Consolidated Framework for Implementation Research guiding the implementation aspects of the proposed work. I have also made this application now under the ACAF scheme, which better reflects my career to date and my future career trajectory. To this end, I will receive guidance and mentorship in my practice development from Professor David Taylor, a leading Pharmacy clinical academic.

Contextual Factors

Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding, or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

5. Plain English Summary of Research

Background

Blood thinning medicines (anticoagulants) are used to treat blood clots. Unfractionated heparin (UFH) is a blood thinning medication. UFH is given to patients by an intravenous (within a vein) drip, particularly in hospitalised patients who are critically unwell, those with kidney disease and patients who undergo heart surgery. UFH offers advantages over other blood thinners - its effects wear off quickly once the intravenous drip is stopped. UFH requires frequent monitoring (up to every 4 hours) to understand if it is working. This is through a blood test. Changes in the UFH dose is guided by the blood test results.

In the UK and around the world, use of UFH can cause serious bleeding problems and even death. The way UFH is dosed and monitored are thought to contribute to this.

We now know the older method of working out the best UFH dose based on a specific blood test, is not ideal. There is now a more accurate blood test and more up-to-date methods to work out the best UFH dosage. These are known as Population Pharmacokinetic Modelling. In my Fellowship I will use the newer blood test and method to work-out how best to dose and monitor UFH.

Aim

To develop a UFH dosing and monitoring decision chart (protocol) using Population Pharmacokinetic Modelling that can be safely implemented in hospitals, ensuring that patients requiring this high-risk medication will be safer and better carer for.

Method

I will conduct three studies:

Study 1 – I will analyse UFH doses and blood test results from patients who received UFH via a drip over a 3-year period at multiple NHS hospitals. Using the Population Pharmacokinetic Modelling method, I will work out the best way to dose and monitor patients with UFH.

Study 2 – using the National Reporting and Learning System which records all adverse incidents from hospitals in England and Wales, I will identify and analyse those related to UFH over a 10-year period. This will help to identify and learn from common problems associated with the use of UFH. This information will help develop a new UFH protocol.

Study 3 - using the results from studies 1&2, with a team of nurses, doctors, pharmacists, carers, members of the public, patients, porters and laboratory staff, I will co-develop a UFH protocol. I will then implement this in multiple hospitals using the principles of implementation science, a field of science that addresses the adoption of evidence-based good practice into clinical practice. I will then study the success of this implementation.

Patient and Public Involvement (PPI)

Patients and members of the public have inputted into the research planned and confirmed its importance and their views taken on board. Through PPI events, I have established a PPI panel who will co-develop the UFH protocol. This panel will also help co-produce patient facing materials and raise awareness and information on why, when and how UFH is used for patients, the benefits of UFH and the importance of the new protocol to benefit patient outcomes. I will be evaluating how PPI has worked at the end of my fellowship.

Dissemination

I will:

-establish a study website where patients, members of the public and health professionals can see how the studies are progressing and what is found and developed and how it is implemented

-publish the results in medical journals and present the findings at conferences

-work closely with organisations such as Thrombosis UK, the British Society of Haematology, and the Anticoagulation Forum, to ensure widespread dissemination and uptake of the resultant work in the UK and across

the world.

6. Scientific Abstract

Background

Unfractionated heparin (UFH) is an intravenous anticoagulant on the World Health Organisation model list of essential medicines. UFH is prescribed to prevent and treat thrombosis. UFH is a leading cause of anticoagulation-related adverse drug events in the UK and health systems around the world. In England and Wales, over the period 1/1/2014-15/11/2022, 603 adverse incidents attributed to UFH were reported to the National Reporting and Learning System. The risks associated with UFH are due to a fine balance between efficacy and toxicity. Therefore, UFH infusions require regular (up to 4 hourly) monitoring through a blood test, the Activated Partial Thromboplastin Time (APTT). However, the APTT test is not a reliable test for UFH and was used historically to inform the dosing schedules for UFH.

There is now (1) a more reliable test, the anti-Xa assay, which better reflects how patients are responding to UFH treatment, and (2) there are better methods for working out UFH drug dosing, using Population Pharmacokinetic Modelling. There is therefore an urgent need to update the way UFH is dosed and monitored in clinical practice, so patient care is improved.

Aim

To develop an IV UFH infusion dosing and monitoring protocol based on the Population Pharmacokinetic Modelling that is implemented in hospitals.

Methods

I will conduct a three-study programme of research. In study 1, using data from hospital records from patients administered an UFH infusion, I will develop a population pharmacokinetic model for UFH. Following multi-centre model validation, I will conduct simulations from the final model to determine the optimal dosing strategy to achieve the desired UFH therapeutic range.

In study 2, I will conduct a retrospective study of all UFH incidents reported to the National Reporting and Learning System over a 10-year period to identify the number and nature of the adverse events relating to UFH, including system failures and defences breached in each incident. This analysis will identify common problems and themes associated with the use of UFH use. The analysis and the lessons from these events will help inform the development of a contemporary UFH protocol.

In study 3, using the findings from studies 1 and 2, with input from patients, members of the public, carers, doctors, nurses, pharmacist's, laboratory staff and porters, I will co-develop a contemporary UFH protocol. I will then conduct an assessment of acceptability and feasibility during implementation into routine clinical practice guided by the Consolidated Framework for Implementation Research.

Impact/Dissemination

My research will lead to a step-change in how UFH is dosed and monitored, leading to effective and safer care. My work has translatability beyond the UK. I anticipate the protocol will become the standard dosing and monitoring protocol for all UFH infusions. Findings will be disseminated to clinicians/patients/public and policy makers, ensuring immediate clinical relevance and translation for patient benefit. Dissemination will be through the website I establish, through social media and organisations such as the British Society of Haematology, Thrombosis UK, the Anticoagulation Forum and the International Society of Thrombosis and Haemostasis.

7. Detailed Research Plan

1. What is the problem being addressed?

Unfractionated heparin (UFH) was the first anticoagulant to become available in clinical practice for the management of thrombosis. One in 4 people die from conditions caused by thrombosis. UFH is included on the WHO list of essential medications as one of the most critical and effective medications necessary to meet the needs of a healthcare system.[1] UFH requires administration as a continuous intravenous infusion and remains the anticoagulant agent of choice in select, often high-risk patients, i.e. those in critical care, patients who undergo cardiac surgery, and those in renal failure.

However, UFH is a leading cause of anticoagulation related adverse drug events in the UK and health systems around the world. UFH is classed as one of 6 drugs most frequently involved in serious medication events[2, 3] and is classified by the Institute for Safe Medication Practices as a "high-alert" medication.[4] Of all records submitted to the American National Error database 2.1% were attributed to UFH, with 5.5% of these incidents causing patient harm.[5] In the National Reporting and Learning System database for England and Wales, from 1/1/2014 to 15/11/2022, 603 adverse incidents including deaths were attributed to UFH.

Despite guidelines, adverse incidents attributed to UFH remain common.[6] Indeed, in my hospital, there were three deaths as a direct result of issues with intravenous UFH within a 6-month period in 2019-2020. Dosing and monitoring are frequently cited as significant factors contributing to UFH adverse events. This stems in part from the fine line that exists between UFH efficacy and toxicity.

To avoid adverse events and ensure therapeutic dosing, UFH requires regular monitoring via blood testing. Additionally, an intravenous UFH infusion involves a number of tasks for multiple members of the clinical team, which makes administration complex and is associated with increased risk and workload. A failure mode and effects analysis study[7] has mapped and identified 233 potential failures and 737 unique potential causes, associated with the use of an intravenous UFH infusion, 45 of which were classified as significant.

Importantly, the original dosing schedules and the monitoring blood test for UFH dosing are now both out of date yet still in use in clinical practice. There is now a blood test, the anti-Xa assay, which better reflects how patients are responding to UFH treatment. There are also better methods for working out drug dosing, known as Population Pharmacokinetic Modelling (PPM). As a result, the current UFH dosing schedule is in need of revision to improve dosing accuracy and monitoring and thus patient safety and outcomes.

2. Why is this research important?

In England, medication errors are estimated to cost £98.5 million, account for 181,626 bed-days, and contribute to 1708 hospital deaths every year.[8] Medication-related errors are thought to cost an estimated 42 billion U.S. dollars worldwide every year.[3]

In 2017, the WHO launched the Global Patient Safety Challenge: Medication Without Harm,[9] This initiative focuses on 4 medication safety domains: (i) patients and the public, (ii) medicines, (iii) health care professionals and (iv) systems and practices of medication management, with 3 priority areas: "polypharmacy," "high-risk situations," and "transitions of care." UFH is a significant contributor to medication adverse-events and falls under domains (ii), (iii) and (iv) and the high-risk situation priority.

Through the development and implementation of a contemporary UFH dosing protocol, I anticipate the number of UFH incidents will be minimised, patient safety will be improved, staff workload will be reduced resulting in improved patient outcomes and care efficiencies.

Preparatory PPI work for my fellowship proposal showed that patients and members of the public have positive attitudes about engaging in medication safety. When seeking guidance about my fellowship topic, patients stated that this is an important area for research including optimising implementation into practice, so there is translation of findings for patient benefit. Patients and members of the public also recommended that they are actively involved in the development of a new UFH protocol.

3. Review of existing evidence

UFH was introduced into clinical practice >50 years ago and despite the availability of newer anticoagulants, remains the drug of choice in hospitalised patients who are critically ill, those undergoing cardiac surgery, and renal patients. This is due to its short half-life and favourable profile in renal failure.[10]

UFH exhibits a marked variation in anticoagulant response among individuals and therefore requires monitoring to ensure adequate anticoagulation (efficacy), whilst minimising the risk of bleeding (toxicity).[11-13] This is achieved using a biomarker as a proxy for plasma concentration. Historically, this biomarker has been the activated partial thromboplastin time (APTT), however APTT is now widely recognised as a sub-optimal biomarker for UFH.[14, 15] In recent years, the anti-Xa assay, calibrated for UFH is replacing the APTT as the monitoring test for UFH. Research shows that the time to reach the therapeutic range and number of subsequent dose changes required are less with anti-Xa compared to APTT monitoring.[16-19] A retrospective cohort study of nearly 20 000 patients concluded that cases monitored by anti-Xa were less likely to require a blood transfusion than patients monitored by APTT after controlling for age, sex and invasive procedures.[20]

A weight-based UFH dosing schedule is widely recommended and used in clinical practice.[21] However, this dosing schedule was developed when modern pharmacokinetic techniques were not available and used the historical biomarker test (APTT).[14, 15] Research shows that when this weight-based dosing schedule is followed only 50% of patients are in the therapeutic range within 24 hours.[22, 23] Other factors, such as age and sex could have important implications for the dosing of UFH.[24-26]

Population Pharmacokinetic Modelling (PPM) provides an opportunity to address this dosing problem. PPM is the science of developing and applying mathematical and statistical models to characterise, understand and predict a drug's pharmacokinetic, pharmacodynamic and biomarker outcome behaviour.[27, 28] Following data collection from the target population, the PPM method develops a pharmacokinetic model to simulate the optimal dose or dosing regimen of the drug in question in the target population. The PPM methodology is endorsed by the European Medicines Agency and the U.S Food and Drug Administration during the drug development process and is additionally applied by clinicians, to address important drug dosing issues at the clinical coalface.[29, 30] To date 3 PPM studies have been published focused on UFH,[31-33] none of which are based on the anti-Xa assay, making my proposal novel.

Despite advances in anticoagulation therapy in the last 10 years, UFH remains the drug of choice in hospitalised patients in select, high-risk hospitalised patient groups. Indeed its use saw a resurgence during the COVID pandemic.[34] However UFH remains a medication associated with patient harm and its dosing and monitoring underpin this. Using the anti-Xa assay and modern pharmacokinetic methods, it is important to revisit the dosing and monitoring approach taken with UFH, in order to improve patient safety and outcomes.

4. Aim and objectives

Aim
To develop a contemporary intravenous UFH infusion dosing and monitoring protocol based on anti-Xa monitoring and population pharmacokinetic modelling and evaluate its implementation into clinical practice.

Objectives
- To characterise the pharmacokinetics of intravenous UFH using PPM
- To analyse commonly reported problems associated with the use of intravenous UFH using the National Reporting and Learning System to inform the UFH protocol
- To co-devlope (with patients and clinicians) and to implement a contemporary UFH dosing and monitoring protocol across King's College Hospital, Guy's and St.Thomas' Hospitals and Royal Brompton and Harefield Hospitals.

5. Research plan

Study 1 – *Population pharmacokinetics of UFH in adults*

Research question: What are the population pharmacokinetics of UFH in adults?

I will conduct a multi-centre PPM study of UFH.

Study setting
I will collect data on patients commenced on an UFH infusion at King's College Hospital (KCH), Guy's and St. Thomas' Hospitals (GSTT), Royal Brompton and Harefield Hospitals (RBHH) – which comprise 6 hospital sites in total. These organisations serve diverse patient populations across London and the Southeast. Collectively, these

organisations house a major trauma centre, a regional kidney disease centre and the largest UK specialist heart and lung centre.

Patient population
I will compile two datasets, a model building dataset (KCH) and a model validation dataset (GSTT & RBHH).

Inclusion criteria
For model development, I will collect retrospective data on all adult patients commenced on an UFH infusion at KCH with anti-Xa monitoring. Patients will be identified through laboratory data of anti-Xa (UFH) assay requests, over the time period 13/10/2021 – 12/10/2024.
From 1/11/2024 to 31/07/2025, I will prospectively identify all new patients commencing UFH at GSTT and RBHH and collect their UFH infusion data. This data will comprise the model validation dataset.
I will exclude patients <18 years, and those prescribed UFH during extra-corporeal membrane oxygenation from both datasets.

Data Collection
I will collect data on the time course of the UFH infusion along with associated anti-Xa concentrations, dosage and subsequent dose changes, until the conclusion of the infusion. I will use the anti-Xa UFH concentration as a surrogate marker for profiling and modelling UFH's pharmacokinetic profile. I have already tested the feasibility of my data collection methods through the completion of two safety audits, at KCH in 2022 and 2023.
I will collect the following variables: age, gender, ethnicity, indication for UFH, weight, height, serum creatinine, potassium, urea, albumin, bilirubin, alkaline phosphatase, haemoglobin, platelet count and coagulation screen at UFH infusion initiation.

Sample size and analysis
To date, three PPM studies have been published on UFH,[31-33] with none using the anti-Xa assay. The sample sizes of these previous PPM studies are: 50 (using activated clotting time), 64 (using APTT) and 41 (using a factor-IIa assay). The two safety audits I conducted at KCH, identified 290 patients over a 1 year period who would be eligible for the present study.
Considering the timeframe outlined of my fellowship proposal, I anticipate at least 870 patients contributing data into the model-building dataset. Assuming each patient contributes 3 anti-Xa readings (minimum) there will be 2610 anti-Xa data points for PPM modelling purposes. This significantly exceeds any previous work in this field.
For the purposes of model validation, I anticipate prospective data on at least 435 patients over the 9 month period from GSTT and RBHH.

I will conduct PPM using NONMEM® software programme. This takes the anti-Xa results collected from the patient population to estimate primary pharmacokinetic parameters.

Specifically, I will explore several structural base models, and will select the model which best fits the data. Goodness of fit plots, a statistical improvement in the fit of the model to the data using the objective function (minus twice the log-likelihood of the data), assessment of the precision of the parameter estimates and residual variability will be the criterion used to evaluate and choose the base model to take forward for full covariate analysis.
Next, I will conduct a covariate analysis, choosing covariates which have a mechanistic meaning, e.g. age, ethnicity, weight. This analysis will involve a classical graphical approach, plotting individual estimates of the pharmacokinetic parameter's random effects against each of the covariates being considered for model inclusion. Each selected covariate will be tested by univariate addition into the base model to confirm relevance. A decrease in objective function value of at least 6.64 ($p<0.01$), will be required to retain the covariate in the intermediate model.
All significant covariates will then simultaneously be added to the base model and their continued relevance will be evaluated using a stepwise backward elimination method, where each covariate will be removed singularly. An increase in the objective function value of greater than 10.82 ($p<0.001$) will be required to retain the covariate in the final model.
Both the base and the final models will be evaluated using a nonparametric bootstrap (1000 replicates). The final model will additionally be evaluated using a visual predictive check, where the 5th, 50th and 95th prediction intervals, simulated from the posterior distribution of the final model parameter estimates will be overlaid with the 5th, 50th and 95th percentiles from the observed data. A well performing model would see the observed percentiles and simulated prediction intervals superimposed.

Model validation
I will then use the prospective model validation data set (GSTT and RBHH) to evaluate the predictive performance of the model developed and thus its effectiveness. No additional fitting will be done. The diagnostic plots and a visual predictive check will be generated as described in the model development section. Model-predicted target concentrations will be computed using the developed model as a prior and information from the validation dataset

only. These predictions will be compared with the observed concentrations by calculating the prediction error, the mean precision error and the root mean square error.
Finally, simulations from the final UFH PK model developed will be conducted to determine the optimal dosing strategy which get patients into the desired therapeutic range.

Study 2 - *Incidence and analysis of UFH adverse events reported to the National Reporting and Learning System*

Research question: What are the number and nature of adverse events associated with the use of intravenous UFH, which can inform the development and implementation of a contemporary UFH dosing and monitoring protocol into clinical practice?

I will conduct a retrospective study of all UFH incidents reported to the National Reporting and Learning System (NRLS).

Data source
The NRLS is a unique resource of patient safety incidents from England and Wales, allowing scrutiny of incidents that have occurred in the National Health Service, so improvements are made and fed back into clinical practice. All NHS organisations and staff are encouraged to report to the NRLS any occurrence of harm or potential harm to a patient as a 'patient safety incident', defined as 'any unintended or unexpected incident which could have, or did, lead to harm for one or more patients receiving NHS care' since 2002. Each NRLS incident report includes structured information: demographic and administrative; the circumstances of occurrence; a categorisation of causation; an assessment of the degree of actual harm; and action taken, or planned, to investigate or prevent a recurrence. Additionally, a section of the report allows the reporting staff member to state in free text, what happened and why they think it happened. Data are anonymised to remove details that identify any patients or staff.

I will request a report from NHS England on all incidents, during the period 1/1/2014-31/12/2023 which include any of the following terms: unfractionated heparin; heparin; heparin AND unfractionated; heparin AND sodium; heparin AND calcium. As part of the request, the following fields will be requested in the output: Incident ID, Care setting, Date, Location, Incident category, Description of what happened, Actions preventing reoccurrence, Apparent causes, Speciality, Reported degree of harm, Source of notification, Medication process, Medication error category, Approved name, Proprietary name, Year of occurrence, Month of occurrence, Staff type, Staff status, Reporting year.

Feasibility: over the period 1/1/2014-15/11/2022, 603 adverse incidents were identified in the NRLS database, when searched using the term 'unfractionated heparin' (personal communication with NHS England). NHS England have confirmed that they will provide the report for the purposes of my fellowship.

Inclusion criteria
All reports outputted over the period outlined, with the search terms described, relating to the use of an intravenous UFH.

Analysis
The setting (adult/paediatric), ward type (e.g. ICU, general medicine), degree of harm reported, and the year in which the incident occurred will be described.
Furthermore, the principal stage of the medication use process in which the patient safety incident occurred, will be categorised according to the following: prescribing, preparation, administration, monitoring, dose adjustment, documentation.[35] Each incident will additionally be categorised according to the NCC MERP index for categorising medication errors.[36] These will be done in addition to the pre-specified NRLS categorisation, to assess whether the detail provided by the NRLS reports allows a more granular categorisation of the events recorded.
Finally, where sufficient detail has been provided, the root cause of each incident will be analysed using the 'Fishbone' methodology,[37] aiming to identify where possible, the specific system failures in each case.

The findings and lessons from this analysis will feed into the development and implementation of the contemporary UFH protocol.

Study 3 – *Co-develop, implement and evaluate the contemporary UFH protocol*

Aim: To co-develop, implement and evaluate the contemporary UFH protocol at KCH, GSTT and RBHH.

Rationale and underpinning theory
The theory guiding the implementation and evaluation components of my fellowship is the Consolidated Framework for Implementation Research (CFIR).[38] As a determinant framework, CFIR provides an appropriate means to predict and/or explain barriers/facilitators to implementation effectiveness. The CFIR constructs reflect factors most likely to influence intervention implementation, with a particular focus on context, and therefore uptake. The CFIR is flexible, and the framework can be tailored to the specific intervention design, factors, and context being studied.[38]

A logic model for study 3 is outlined in **figure 1**.

Preparatory work
Initially, I will apply the CFIR framework to each hospital site, to map the internal and external factors that might influence implementation of the UFH protocol and to inform the data collected.
This will identify key stakeholders at each site driving the use of UFH, who I will then invite to form a heparin stakeholder group across the study sites.
The stakeholder group will be multidisciplinary: patients, carers, doctors, nurses, pharmacists, porters (involved in sample transportation) and laboratory staff.

This stakeholder group will meet 12 times over years 2 and 3 of my fellowship and will co-develop and implement the UFH at the study sites.

Study 3 will be completed in 3 phases.

Phase I - understanding needs and context

To contextualise the use of UFH at the participating hospitals, I will address the following objectives:

1. To compare and contrast the UFH protocols currently in place at each hospital site
2. To observe how UFH infusions are currently used in practice, including monitoring practices and understand the perceived problems with UFH infusions experienced by front-line staff
3. To explore the nature of UFH incidents reported at the participating hospitals

Objective 1
I will obtain current UFH protocols in operation at each hospital site and analyse their content. Components will be categorised into (1)indications for which UFH is suggested, (2)recommended UFH dosing, (3)how the UFH infusion is set-up and administered, (4)target anti-Xa concentrations, (5)suggested changes in the dose rate according to results, (6)frequency of monitoring, (7)how samples are processed and reported by the laboratory and (8)where staff can access expert help.

This content will be entered into a data collection tool in Microsoft Excel and analysed for similarities and differences between hospitals.

Output: A descriptive map[39] of what commonality/differences exist in UFH protocols across the sites.

Objective 2
I will conduct a multi-site qualitative *go-long* interview study,[40] which will focus on how clinical staff deliver UFH infusions in practice and what barriers/enablers there would be to a new UFH protocol.

Participants: I will purposively sample front-line clinical staff who are involved in the delivery of the UFH infusions. Data collection will combine observations of UFH delivery in practice with conversations, making it possible to observe spatial practices *in-situ* while gathering insights from the interviewee. I will conduct 24 *go-along* interviews (3 nurses, 2 doctors, 1 pharmacist, 1 lab staff, and 1 porter from each hospital).

Incentives: £25 e-gift card provided to each volunteer.

Data collection: I will develop a topic guide, in consultation with the stakeholder group and guided by CFIR.

Data analysis: I will follow principles of deductive reasoning.[41] This will involve an iterative dialogue whereby preliminary lines of data interpretation are tested with reference to existing theory to confirm or to extent explanatory reasoning. I will relate back codes and themes identified to the domains of CFIR.

Output: An understanding of what issues are faced by the front-line staff in delivering UFH infusions safely and how receptive they are to a new UFH protocol.

Objective 3
To build a picture of the specific nature of UFH adverse events occurring at each hospital, I will interrogate the error reporting system at each site to identify all incidents linked to UFH at each hospital over the period 01/2022 – 12/2024.

The data analysis for this will follow that previously described in study 2.

Output: This will complement the findings from study 2 of the NRLS analysis and provide an insight into any particular problems associated with UFH at each hospital.

Following completion of phase I, coupled with the findings from studies 1 and 2, a contemporary UFH infusion protocol will be co-developed by the stakeholder group and ratified at KCH. The protocol will be launched at KCH, through a planned implementation.

Phase II – pilot and assess

Following implementation at one of my study sites, I will evaluate acceptability, fidelity, safety and appropriateness 3 months into implementation.

Acceptability
I will conduct a cross-sectional study asking staff involved in the delivery of UFH protocols to complete the Theoretical Framework of Acceptability questionnaire.[42]

Sampling: I will purposively sample 50 members of staff (nurses, doctors, pharmacists, porters, and laboratory staff) at the study site in cardiology, ICU and renal specialities.

Incentives: £15 e-gift card for completion of questionnaire

Analysis: The Theoretical Framework of Acceptability questionnaire comprises 7 constructs: affective attitude, burden, ethicality, perceived effectiveness, intervention coherence, self-efficacy and opportunity costs. There is also a single question around general acceptability. The scales are on a 5-point Likert scale, with the negative extreme descriptor scored at 1. A single acceptability score will be computed generating a mean score of the 7 constructs. Items assessing burden and opportunity costs will be reverse scored (higher score indicates higher acceptability). I will correlate this score with the overall acceptability item. I will use descriptive statistics to represent the findings.

Output: An indication of how acceptable the contemporary UFH protocol is to front-line staff at the pilot site.

Fidelity
Underpinned by the Conceptual Framework for Implementation Fidelity,[43] a fidelity checklist will be compiled and agreed by the UFH stakeholder group. This checklist will primarily assess whether staff adhere to the (core and non-core) aspects of the protocol, as defined by the stakeholder group.

Data collection: I will assess fidelity by retrospectively reviewing all UFH infusions administered between weeks 4 and 11 at KCH during the pilot period, by reviewing the electronic medical / prescribing / laboratory records using the compiled checklist.

Analysis: The checklist will have components relating to the prescribing, preparation, administration, monitoring, reporting of results and adjustments of the UFH infusion. Scoring of the fidelity checklist will be computed by selections of ‘Yes’, ‘No’ and ‘Nondelivered’. A total score will be generated based on the number of delivered components divided by the number of components that were deliverable and multiplied by 100 to give a % fidelity score.

Output: An assessment of the degree to which the UFH protocol is delivered as intended.

Safety and Appropriateness

I will evaluate appropriateness of the contemporary UFH protocol through prescribing data and safety through any incident reports which emerge during the implementation.

Data collection: I will identify all UFH infusion prescriptions through the e-prescribing system at KCH, retrospectively and I will assess:

-% of patients reaching target anti-Xa concentrations within 6 and 24 hours of commencing the infusion when the dosing protocol is followed
-% of patients overshooting (>1 IU/mL) or undershooting (<0.1 IU/mL) when the dosing protocol is followed

Through the incident report systems in the hospitals, the number and nature of any UFH adverse incidents reported will be collated and investigated.

Analysis: Descriptive statistics will be used to report findings. The number and nature of any safety incidents will be described.

Output: An indication of how well the dosing protocol is performing when followed.

The findings from the acceptability, fidelity, safety and appropriateness studies will be triangulated and discussed with the stakeholder group to inform what, if any, refinements should be made to the UFH protocol, before ratifying and launching at GSTT and RBHH.

Phase III – full implementation

The acceptability, fidelity, safety and appropriateness assessments, as outlined in phase 2, will be completed across the 3 hospitals - 3 months post implementation for GSTT and RBHH and 6 months for KCH.

As well as generating data for each specific hospital, differences in the findings between hospitals and sites will be compared.

Following the conclusion of my fellowship, fidelity and appropriateness will be re-assessed at each hospital, 1 year following implementation. These assessments will follow the methodology as described previously and conducted by either my undergraduate or masters students at KCL, following appropriate training.

A key aim of my fellowship is to have a standardised UFH protocol across the health service. Therefore, penetration of the protocol into the NHS will be studied 3 years following the implementation of the protocol at KCH, GSTT and RBHH. My research students will conduct this, along with fidelity, safety, appropriateness and acceptability studies, under my supervision.

6. Dissemination, Outputs and anticipated Impact

Outputs
My fellowship will lead to the development and implementation of a contemporary UFH dosing protocol. There will be significant publications from my fellowship focused on: (i) PPM work of UFH, (ii) analysis and learning from past UFH adverse events reported to NRLS, and (iii) implementation and evaluation of the UFH protocol into practice.

Dissemination
Early on in my fellowship, I will establish a website, which will provide live information on the progress of my work to members of the public, patients and healthcare professionals, so there is transparency on how the UFH protocol has been developed and what steps are being taken to implement the findings.
My PPI group will advise and co-develop the patient and members of the public material for the public areas of the website, as well as advise and co-develop the materials disseminated through the Thrombosis UK website, Facebook and X accounts.
The website will have a clinician specific section, which will update healthcare professionals on the programme of work and what results are emerging. The transparency of this will lead to greater penetration of the final UFH protocol into clinical practice, once implemented.
Throughout my fellowship and beyond, the findings will be disseminated as oral presentations and posters in real-time at appropriate conferences, e.g. ISTH, so that the results undergo wide peer review and be scrutinised. The

findings will also be published in peer review journals.
I am a member of the BSH guideline writing group for Measurement of Heparin and other non-Coumarin anticoagulants and their effects on tests of Haemostasis. I am therefore well placed to ensure the work from my fellowship informs the BSH update of the UFH guideline.

Impact
Through completing the work outlined in my fellowship, patients will receive safer and more effective care. The UFH dosing protocol developed will be available for centres to use and through the due diligence applied in its development, will be universally applicable across the health service in the UK and around the world.

7. Project management

I will be responsible for project management and completion of the work proposed. With the support of my research support mentors, we will meet every 4-6 weeks as a group, to discuss progress and I will take on-board advice and guidance provided. I have extensive experience of seeing research studies to completion and I am confident that this will be the case here.

Additionally, during year 2 of my fellowship I will form a heparin stakeholder group formed from the multiple centres taking part, who will meet regularly during year 2 and 3 and co-develop and implement the UFH protocol in their respective organisations.

My PPI panel will additionally meet every 6 months to specifically discuss work plans relating to PPI in my fellowship which will be chaired by a constitute member of the group. This panel will also be part of the wider heparin stakeholder group.

8. Ethics

Ethics approval and governance approvals will be sought, prior to commencing studies 1, 2 and 3. In order to commence work soon into the fellowship, ethics approval for study 1 will be sought prior to the start of the fellowship.

9. Success criteria

(i) a Population Pharmacokinetic model of UFH in adult patients
(ii) an understanding of number and nature of what has gone wrong in the past with UFH infusions and learning lessons from these incidents
(iii) co-developing and implementing a contemporary UFH protocol into clinical practice

The risks identified are associated with success criteria (i) and (iii). The retrospective nature of the model development dataset means that data could be missing. However, this has been mitigated, by already completing 2 safety audits at KCH, which demonstrates that all the data required for PPM purposes is present and obtainable. This is the same for GSTT and RBHH as well.

Implementation of anything new is challenging and indeed is an element of the work that is a key development target for me to learn about, as part of my fellowship. There is a risk that the UFH dosing protocol is not taken-up into clinical practice. To mitigate this, the CFIR is guiding the implementation and a stakeholder group formed from the 3 organisations will co-develop and implement the protocol. This will ensure ‘buy-in’ to the final protocol and will be fully supportive of its implementation and facilitate this into clinical practice. The results from all 3 studies will be made widely available, so others can independently scrutinise the results and conclusions drawn, which should ensure there is wide penetration of the final protocol.

8. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I convened two PPI events to inform the research outlined in my fellowship. One comprised 3 patients who were familiar with heparin, including a representative from the charity Thrombosis UK and the second group comprised 3 members of the public.

Following a presentation on the background and what work was being proposed, both groups confirmed that the proposed research was an important area to tackle and particularly liked the idea of using healthcare data to address the research questions of studies 1 and 2, with the members of the public group stating it made good economic sense. Both groups also commented that if anonymised data is used for something like this and helps the NHS then makes a lot of sense to do. The anticoagulated group confirmed that learning from past mistakes and analysing information from the National Learning and Reporting System was an important component in the work proposed.

The PPI groups commented on the fact that research implementation was not something they had thought about before but said this was crucial and "…makes total sense that you spend time taking the findings and implementing into clinical practice." Some concluding remarks from the groups: "Very worthwhile project." "Why hasn't this been done already!?"

Following PPI review, the following changes were made to research proposal:

-making clearer what will be learnt from study 2 and how that can be applied to the final protocol developed

-ensuring that nurses, doctors, pharmacists, porters and laboratory staff are involved the UFH heparin protocol development. And to actively involve patients and members of the public in the process.

The 2 PPI events additionally confirmed that the public/patient voice in the work is important, and the importance of continued involvement was emphasised; which has now led to a PPI panel being formed with representation from Thrombosis UK to continue to input into my fellowship and be PPI stakeholders in the development of the UFH protocol and as research champions.

Please describe the ways in which patients/service users and the public will be actively involved in the proposed research, including any training and support provided

Following a recommendation from the preparatory PPI work, a panel comprising of patients and members of the public has been formed. This panel comprises 2 patients who have previously received UFH and 2 members of the public and will be actively involved in the development of the UFH protocol.
The panel, chaired by a constituent member, will be supported with their PPI activities through meetings held every 6 months throughout the lifecycle of my fellowship. These meetings will provide an opportunity to get updates on the work and impending work streams.

The panel will also lead and co-develop the materials for public dissemination, from the start of the proposed work outlining to patients and the public why the work is being completed, to when results begin to emerge. This will primarily be led through the public section of the website that will be established, ensuring that the information and materials that are posted there is communicated in plain English. The panel will also provide advice on how best to liaise with organisations like Thrombosis UK and will co-develop the material posted on Thrombosis UK communication channels (Facebook, X and website), so there is wide-spread dissemination through Thrombosis UK communication channels, as well.
The panel will also attend and contribute to the unfractionated heparin stakeholder meetings and co-develop the UFH protocol. Finally, the findings from the implementation work will be presented at the Quality and Safety in Healthcare conference. The chair of the PPI panel will present the findings with me at this conference.

To enable the PPI panel members, following a needs assessment, training will be provided through the Imperial College London public involvement training aimed at patients and members of the public. The panel will be recompensed for their time for undertaking this role, as per NIHR guidelines. At the conclusion of my fellowship, the involvement and impact of PPI involvement will be evaluated through a workshop undertaking a cube framework exercise, the findings of which will also be published, so any learning can benefit the wider research community.

If it is considered not appropriate and meaningful to actively involve patients/services users, carers and the public in your proposed research, please justify why
Not applicable.

9. Training & Development and Research Support

Proposed training and development programme

Gap analysis

In order to establish myself as an international research leader, a gap analysis was conducted. Using the vitae researcher development framework, the following were identified under the domain of personal effectiveness:

i. Has national, international and policy-making network connections with academic and non-academic bodies and organisations.
ii. Has influential connections with significant bodies and organisations.

In addition, I have identified that I have little knowledge of the field of implementation science and the field of epidemiology, which I will formally study during my fellowship.

Through completing the work proposed in my fellowship, the step-change in my career will occur, which will establish me as an international leader and authority on anticoagulation therapy and with that, have exceptional influence, with input sought from policy makers, as befits an internationally recognised researcher.

Research support

Suitable research support mentors for each stage of the research proposed in the application have been identified, who would not only enhance the work undertaken but are international research leaders in their own fields and prefect role-models for my transition to this stage. The selection of each has been carefully considered, with differing hierarchical distance from one another, and differing backgrounds and training, which will be complimentary for my fellowship.

My research support mentors will meet with me, as a group every 4-6 weeks to advise, mentor and support. Depending on the stage of research being undertaken, each research mentor will take a lead role during my fellowship, complimenting their specific expertise. Each research mentor has agreed that I can additionally contact them on an ad-hoc basis for research or personal developmental advice, in their roles as international research leaders.

Practice development

As a consultant pharmacist of >10 years, my practice in anticoagulation practice is already of a high standard. Therefore, for practice development during my fellowship I will maintain my clinical practice currency, through inpatient ward-rounds and outpatients anticoagulation clinics at KCH, whilst I undertake the research. I have additionally and importantly invited Prof Taylor to be my practice development mentor. As a leading pharmacy clinical academic, Prof Taylor has translated his practice research to international clinical practice, which is what I wish to emulate in the anticoagulation field. I will meet Prof Taylor every 2 months to practice mentor me and advise on how to develop my international presence to translate my practice-research findings internationally.

Additionally, as my fellowship is focused on anticoagulation stewardship, I will undertake a practice visit with Dr Alison Burnett, President of the Anticoagulation Forum (AF) in North America. AF are world leaders on anticoagulation stewardship. I will learn about the initiatives that are in place and bring back learning to the UK. I will also use this visit to foster a relationship with AF, to help with dissemination aspects of my fellowship.

Training

Implementation Science

A key reason for undertaking this fellowship is to formally learn about implementation science and its application in the healthcare setting. I will take the following module at Oxford University: Knowledge into Action.

This particular module was chosen, based on the learning outcomes, the opportunity it provides to work on the UFH problem as part of the assignment that is completed, the opportunity it will provide to interact and network with other healthcare professionals and students tackling implementation projects in their specific fields and the timing of when the module runs. The residential component of the module runs in May of each year. Considering when I plan to start the fellowship, May time of the first year of the fellowship is the perfect time to learn about implementation science, as discussions on implementation will begin as I enter year 2 of the fellowship. This module provides a sound foundation in this field.

The module is run over an eight-week period, using a hybrid teaching method, providing additionally flexibility.

This module incorporates a 1 week residential at Oxford for face-to-face teaching, which will afford me the opportunity to network with others doing implementation work. The assessment for this module, is a 4000-word assignment, which will be anchored on the UFH protocol implementation work. I will also attend the NIHR ARC South London Implementation Science Conference and the International Quality and Safety in Healthcare Conference in the final year of my fellowship, to further network and learn from experts working in the implementation science field, as I implement and evaluate the developed UFH protocol.

Epidemiology

Study 2 has an epidemiology component, which I am not formally trained in. I will therefore undertake 3 distance learning modules from the London School of Hygiene and Tropical Medicine master's programme in epidemiology:

EPM101–Fundamentals in Epidemiology

EPM102–Statistics for Epidemiology

EPM103–Practical Epidemiology

The content of the modules is learnt through self-direction of materials provided and will allow me to complete the material at appropriate times during year 1 of my fellowship, to support study 2, and provide the requisite foundation required in this discipline.

Personal development

Leadership

During my fellowship, I will refine my leadership skills and complete selected e-learning modules, from the NIHR leadership lab modules to address personal developmental objectives and my fellowship proposal:

-Collaborating across organisational boundaries

-Courageous and compelling conversations

-Building resilience to navigate complexity and change

-My leadership brand

-Building strength in my team

-Leading change and innovation

These modules will be undertaken at appropriate times during my fellowship, so the learning can be applied in real-time.

Coaching

As part of my personal effectiveness development, I will undertake a series of coaching sessions in the final year of my fellowship, in the order to reflect and develop the right attributes which would ensure my personal effectiveness on the international stage. The coaching will provide a non-directive form of development, through a series of purposeful conversations which will enable me to move towards achieving my research, practice and personal goals. The coaching is scheduled for the final year of my fellowship, as that is the time, I will be addressing the implementation phases of the research and liaising with organisations in the UK and internationally.

Research Support

RESEARCH SUPPORT 1

Name	Joseph Standing

Institution	UCL

Position	Professor of Pharmacometrics

Description of support provided
Prof Standing holds a PhD in clinical pharmacology, and MSc in Statistics. He is a pharmacist by background and a member of the Royal Pharmaceutical Society, British Pharmacological Society, Fellow of the Royal Statistical Society, member of the European Society for Clinical Microbiology and Infectious Diseases and member of the International Society of Pharmacometrics. Prof Standing has held two MRC academic fellowships (Methodology 2011-2015, Clinician Scientist 2015-2020) which have led to the formation of his pharmacometrics research group at the UCL Great Ormond Street Institute of Child Health. His group is funded from research council and European agencies (MRC, EU-FP7, Swedish Research Council) and charities/non-profit (Wellcome, DNDi, GOSHCC). Prof Standing is the MSc module lead for the Pharmacometrics module at UCL, where students gain theoretical and practical computational knowledge on population PKPD. Prof Standing also teaching pharmacometrics internationally on an ad-hoc basis - recently includes Kenya and Malaysia. Prof Standing has a vibrant research programme, having supervised 9 PhD students to completion and currently supervises 9 PhD students. The academic background of these students ranges from pharmacy, medicine, and biology to physics and engineering. Prof Standing has medication regulatory and industrial collaboration experience, having served 3 years on the European Medicines Agency (EMA) Modelling and Simulation Working Party, been on several EMA expert groups (mainly on paediatrics and infectious diseases), and has current active industry collaborations with AstraZeneca, BenevolentAI and Lilly. Prof Standing is the perfect research support mentor for study 1 of my fellowship.

RESEARCH SUPPORT 2

Name	Bryony Dean Franklin

Institution	UCL & Imperial College Healthcare NHS Trust

Position	Professor of Medication Safety

Description of support provided
Prof Franklin is a hospital pharmacist and Director of the Centre for Medication Safety and Service Quality, a joint research unit between Imperial College Healthcare NHS Trust and UCL School of Pharmacy, where Prof Franklin is Professor of Medication Safety. Prof Franklin is co-director of the NIHR West London Patient Safety Research Collaboration and theme lead for the NIHR Health Protection Research Unit in Healthcare Associated Infection and Antimicrobial Resistance at Imperial College London. Prof Franklin has been involved with patient safety research for more than 25 years; she has published widely on medication safety, the evaluation of various technologies designed to reduce errors, and the patient's role in patient safety. Her current work combines research, quality improvement, education and training, and hospital pharmacy practice. Prof Franklin is Co-Editor-in-Chief for the journal BMJ Quality and Safety. She is also co-editor of the textbook "Safety in Medication Use" and co-author of "Going into Hospital? A guide for patients, carers and families", a book aimed at the general public. Prof Franklin is currently and has in the past undertaken studies using National Reporting and Learning System data. Coupled with her research programme in medication safety, Prof Franklin is the ideal research support mentor for study 2 of my fellowship.

RESEARCH SUPPORT 3

Name	Louise Rose

Institution	King's College London

Position	Professor of Critical Care Nursing

Description of support provided
Prof Rose is a nurse by background and professor of Critical Care Nursing at KCL and an honorary Professor in Critical Care at the Lane Fox Respiratory Unit at St Thomas' Hospital. Prof Rose has published over 205 peer-reviewed publications and authored four book chapters. She has been awarded 106 peer reviewed competitive research grants totalling >£10 million. Prior to joining KCL, Prof Rose was Associate Professor at the University of Toronto and remains the Research Director of the Prolonged Ventilation Weaning Centre there. Prof Rose's research program focuses on improving outcomes and the healthcare experience of mechanically ventilated patients across the care continuum from the emergency department, intensive care unit, specialised weaning centre, and at home. Through her collaborations with specialised weaning centres in both Canada and the UK, Professor Rose has a particular research interest in the delivery of care to long-stay ICU patients and their families, and ICU survivorship. Prof Rose has extensive clinical experience of intensive care in four countries and has set up a Masters in Critical Care Nursing at Addis Ababa University in Ethiopia. During the COVID pandemic, Prof Rose co-founded and implemented Life-Lines, a philanthropic initiative that provided >1,400 4G enabled tablets with a bespoke ICU virtual visiting solution to >180 NHS hospitals. Prof Rose has extensive experience of implementation, particularly in the application of the CFIR, having widely implemented initiatives in the critical care setting internationally and will provide research support in the implementation aspects of my fellowship (study 3).

Practice Support

PRACTICE SUPPORT 1

Name	David Taylor

Institution	South London and the Maudsley Hospital

Position	Director of Pharmacy and Pathology & Professor of Psychopharmacology

How will the proposed practice support arrangements contribute to the development of the applicant?
Prof Taylor, is Director of Pharmacy and Pathology at the Maudsley Hospital and Professor of Psychopharmacology at King's College London, as well as head of the Pharmaceutical Sciences Clinical Academic Group at King's Health Partners. He is the Editor-in-Chief of the journal Therapeutic Advances in Psychopharmacology and the lead author of the Maudsley Prescribing Guidelines, which since their inception in 1993 has sold over 300,000 copies in fourteen editions and twelve languages. In addition Prof Taylor serves on the editorial board of several journals. Prof Taylor has authored over 400 clinical papers with >21K citations and a H-index of 68. As well being a Fellow of the Royal Pharmaceutical Society of Great Britain, Prof Taylor is a fellow of the Royal College of Physicians (Edin) and an Honorary Fellow of the Royal College of Psychiatrists. As a clinical academic, Prof Taylor has built an international reputation in the field of psychiatric medicine based on his clinical practice and research. I have chosen Prof Taylor to mentor me as a practice mentor during my fellowship, as he is the perfect exemplar of an international clinical academic in his field. He maintains his clinical practice with a vibrate research programme. Prof Taylor will guide and mentor me to develop as an international clinical academic leader in the anticoagulation field.

Collaborations

The collaborations I have fostered and developed will ensure success and uptake of the UFH protocol developed.

Already established:

1. Guy's and St. Thomas' Hospital and the Royal Brompton and Harefield Hospitals

Prof Beverley Hunt, lead coagulation haematologist at GSTT and RBHH and the haematology teams based there have agreed to collaborate on the proposal. This will mean the Population Pharmacokinetic Modelling work will be validated in these 2 external centres using prospectively collected data, ensuring the credibility of the final model developed. I will also be implementing the final UFH model developed at these external centres.

2. British Society for Haematology (BSH)

The BSH promotes excellence in the study, research, and practice of haematology. The BSH is responsible for writing UK haematology guidelines, including thrombosis and haemostasis guidelines. The last society guideline on the use and monitoring of heparin[44] was published by the BSH in 2006 and requires updating. The chair of the BSH thrombosis and haemostasis task force, Dr Keith Gomez, has confirmed the importance of the work and I have been invited to be a member of the UFH guideline writing group which will be updating this guidance in the future. This will position me well, for using the work from this fellowship to update this guideline, ensuring widespread uptake and dissemination.

3. Thrombosis UK

This UK charity increases awareness, supports research and extends understanding through education and the sharing of information to improve care for all those affected by thrombosis. Jo Jerrome, chief executive of the charity, is fully supportive of the work proposed. The charity has nominated a patient to be part of the PPI panel for my fellowship work. Thrombosis UK will also help with widely disseminating the work through their communication channels with patients, members of the public, carers and healthcare professionals.

4. The United Kingdom Clinical Pharmacy Association (UKCPA)

An organisation of UK clinical pharmacists, promotes expert practice in medicines management. The organisation provides a perfect platform to reach front-line pharmacists across the UK to share the development of the UFH protocol and promote best practice. The Chair of the Haemostasis and Thrombosis sub-group of the UKCPA has also endorsed the work proposed in the application and will support with developmental and dissemination aspects of the work undertaken.

5. Anticoagulation Forum

An international organisation helping practitioners improve anticoagulation patient care. Its membership includes more than 13,000 healthcare professionals representing over 3,000 anticoagulation services. Anticoagulation Forum has exceptional influence on the practice of anticoagulation. I have already connected with the organisation's president, Dr Allison Burnett, who has agreed to host me and to share best practice in the UFH field and the anticoagulation stewardship work they lead on, in USA. I will use this opportunity to ensure that the work from my fellowship feeds into the work of Anticoagulation Forum and foster this relationship to ensure worldwide dissemination of the resultant UFH work.

Post fellowship:

International Society of Thrombosis and Haemostasis (ISTH)

Is the leading organisation dedicated to the prevention and treatment of thrombosis. The ISTH brings together clinicians from >100 countries, with ISTH guidelines having world-wide impact. The Scientific and Standardisation Committee (SSC) on control of anticoagulation of ISTH, is focused on laboratory and clinical issues related to the use of anticoagulant drugs. During the latter part of my fellowship, I will apply to be a committee member of the control of anticoagulation SSC, which will afford me the opportunity to champion the new protocol in the organisation with the aim of the heparin work informing a ISTH specific guideline from the SSC.

Host Organisation Support Statement

The following statement of support must be completed collaboratively by the heads of department at the Host Organisation (who will administer any award) and the Partner Organisation.

The Heads of department will be able to access this section of the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

The department of haematological medicine at King's College Hospital NHS Foundation Trust (KCHFT) and the Institute of Pharmaceutical Sciences at King's College London (KCL) welcomes the opportunity to host Jignesh's NIHR advanced clinical academic fellowship.

KCHFT is one of the UK's largest and busiest teaching Trusts with King's specialist services available to patients across a wide catchment area, providing nationally and internationally recognised work in liver disease and transplantation, neurosciences, haemato-oncology and fetal medicine. KCHFT houses two hyper-acute stroke units, the largest bone marrow transplant centre in Europe, and is one of London's four trauma centres. In partnership with King's College London (KCL), the Trust has recently been awarded a National Research Centre in Patient Safety and Service Quality.

The haematology department, where Jignesh will be based, is one of the UK's largest, with an extensive referral base. There is a vibrant research culture within the department which has demonstrated continued success in leading recruitment to many national and international clinical trials. Our departments research publishes in high impact peer reviewed journals, e.g. N Engl J Med (2022) 386:11-23; Blood (2022) 139:2642-2652; Blood (2020) 136:1347–1350, demonstrating the extensive and innovative research taking place. We make significant contributions to the ever-evolving body of knowledge and helping advance care and treatment options for our patients. At any one time, we have a number of research students (PhD, MD, MSc students (including international)) and post-doctoral students based in the department. In addition to this, our department currently hosts 3 NIHR/MRC CARP awards, held by 3 consultant haematologists and Jignesh will directly benefit from this exposure during his fellowship. The thrombosis team at King's is a leading multidisciplinary clinical academic group within the department. During the last 10 years, 6 doctors have completed their MD(Res) studies and 4 pharmacists have successfully completed their PhD's within this group, with 4 specifically conducting pharmacokinetic studies as part of their research programmes. During his fellowship, Jignesh will attend the monthly research group meeting, to report on progress and to provide a steer on other studies ongoing within the group.

There is a structured partnership in place between KCHFT and KCL through the Academic Health Science Centre of King's Health Partners. As a clinical academic, Jignesh will benefit from the links at the Institute of Pharmaceutical Science in KCL, where research activity revolves around three strands: i) drug discovery, ii) medicines development and iii) medicines use. In the REF 2021 exercise, KCL's submission for Allied Health Professions, Dentistry, Nursing and Pharmacy (UoA3) was ranked 1st of 90 submissions overall in the UK (by both power and GPA) and 1st for Environment (GPA 4.00), Impacts (GPA 3.86) and for the number of 4* Outputs. Indeed, Jignesh's research featured, as part of the environment statement which was returned under this UoA3.

The Institute of Pharmaceutical Science (IPS) has a strong track record of supporting clinical academics to realise their research ambitions, with >15 clinical pharmacists completing their PhDs within King's Health Partners during the past 10 years. IPS staff supervise over 100 PhD students and monthly PhD student-led research seminars facilitate networking and discussion across our postgraduate research community culminating in the IPS Postgraduate Research Symposium which is a highlight of the year.

Both our departments have clear commitments to championing initiatives and processes that facilitate the progression of staff from underrepresented groups into leadership positions and ensuring that EDI initiatives and principles translate into meaningful and quantifiable positive changes at all levels. Our departments also fully committed to supporting the wellbeing and equality of opportunity for our student / staff community and the delivery of zero tolerance to bullying, harassment and discrimination.

Several initiatives span King's Health Partners which Jignesh will continue to benefit from during the fellowship, e.g. the King's Clinical Academic Training Office (KCATO), providing trainee-centred guidance to navigate the career journey so that King's clinical academics emerge as highly skilled clinicians with the ability to deliver globally impactful research, as well as being inspirational role models. In addition, the Advanced Practice Education & Research Hub, hosted by the Florence Nightingale School of Nursing & Midwifery, will expose Jignesh to a wider network of allied health professionals who are also developing and succeeding in clinical academic careers. Finally, Jignesh will benefit from the world leading seminars that are held across King's Health Partners, where researchers from around the world are invited to present and share their research.

Support for Jignesh

Jignesh is a high-achieving clinical academic in our departments, and widely acknowledged to be a rising star in

the field of thrombosis and anticoagulation. He has a strong academic portfolio, with several important publications in internationally recognised journals to his name. He is a highly organised, collegiate and committed researcher who has established research collaborations across King's and with other groups around the UK and further afield. His research has always been highly patient-centric, dealing with important and unanswered issues in anticoagulation. He was hosted in our departments through his very successful GSTT/KCL BRC PhD fellowship (2009-2012), where he modelled enoxaparin pharmacokinetics during the antenatal period. Hosting Jignesh through this fellowship, where he plans to model unfractionated heparin pharmacokinetics aligns with the departments expertise and capability and the wider King's Health Partners mission to create and nurture the next generation of the NHS clinical academic workforce. The fellowship is also an appropriate next step for Jignesh, in becoming an international research leader in the field of anticoagulation practice and a role model for academic healthcare professionals, particularly pharmacy clinical academics. As an existing member of our departments, Jignesh has made full use of the research support infrastructure in preparation for his application and will continue to receive such support. In the long-term, we have an infrastructure ready to ensure Jignesh achieves his research ambitions which will benefit patients. In our view he would be an ideal recipient for this award.

10. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the Fellowship as described in the proposal.

This section must be completed by the Applicant in conjunction with the Research Support Office or Finance Office at the host organisation.

STAFF DETAILS 1

Staff type	Lead Applicant

Name	Jignesh Patel

Employing organisation	King's College Hospital NHS Foundation Trust

Role on this research	Fellowship holder

Grade	8c

Starting point on scale	Point 46

Increment Date	01/10/2024

Justification of Costs
This cost covers my current salary for the duration of the fellowship.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	81138.70
Geographical Weighting (£)	7746
Other allowances (£)	568.36
Superannuation (£)	12781
Nat. Insurance (£)	12007
Current Annual Costs (£)	**114241.06**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Dr Jignesh Patel
Type of cost: NHS
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Geographical weighting includes ILW and an apprenticeship levy of £568.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	81138	81138	81138	243414
National Insurance	12007	12007	12007	36021
Superannuation	12781	12781	12781	38343
Geographical weighting / Other allowances (if applicable)	8314	8314	8314	24942

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	0	0
Total NHS	342720	0	0	342720
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: Other
Description: Conference Fees
Conference: True
Item Description: Population Approach Group Europe (PAGE) - Conference registration fees
Justification of Cost: The PAGE conference is the key annual conference in pharmacometrics, bringing together experts in this field. Attendance at this conference will allow me to network with these experts and share findings and receive peer review from the PPM work. The conference will be in a European destination. The cost listed is based on registration for the 2023 conference.

	Year 1	Year 2	Year 3	Total Cost (£)
	380	0	0	380

Type of cost: Other
Description: Journey Costs
Conference: True
Item Description: Travel to PAGE 2025 conference
Justification of Cost: This cost covers the return flights travel to the conference at a European destination for the PAGE 2025 conference.

	Year 1	Year 2	Year 3	Total Cost (£)
	250	0	0	250

Type of cost: Other
Description: Subsistence
Conference: True
Item Description: Accommodation and food
Justification of Cost: The PAGE conference is a 4 day conference and this cost covers the cost of accommodation and subsistence.

	Year 1	Year 2	Year 3	Total Cost (£)
	370	0	0	370

Type of cost: Other
Description: Conference Fees
Conference: True
Item Description: International Society of Thrombosis and Haemostasis (ISTH) 2026 registration fees
Justification of Cost: The ISTH is the most important conference in the field of thrombosis and haemostasis. Attendance at this conference will allow me to network with world experts in this field and afford me the opportunity to present findings from the work completed to date. This cost covers the registration fee for attending the conference.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	300	0	300

Type of cost: Other
Description: Journey Costs
Conference: True
Item Description: Travel to ISTH 2026 conference
Justification of Cost: ISTH is an international conference. In 2023, the conference was hosted in Montreal. This cost covers the estimated flight costs to the ISTH conference, based on the 2023 venue.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	700	0	700

Type of cost: Other
Description: Subsistence
Conference: True
Item Description: Accommodation and food
Justification of Cost: This cost covers accommodation and food at the ISTH 2026 conference for 3 days.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	500	0	500

Type of cost: Other
Description: Conference Fees
Conference: True
Item Description: 2 day pass
Justification of Cost: This cost covers attendance (2 day pass) at the International Quality and Safety Forum conference in 2027. The costs are based on registration fees for the 2024 conference and I plan to present the implementation evaluation work when attending at this conference and use the opportunity to network and learn from experts in the field of implementation science.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1070	1070

Type of cost: Other
Description: Journey Costs
Conference: True
Item Description: Travel to International Forum on Quality and Safety in Healthcare
Justification of Cost: This cost covers the return journey costs to the conference.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	580	580

Type of cost: Other
Description: Subsistence
Conference: True
Item Description: Accommodation and food
Justification of Cost: This costs covers a 2 night stay and subsistence cost at the international conference.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	350	350

Type of cost: Other
Description: Conference Fees
Conference: True
Item Description: South London NIHR ARC Implementation Science conference
Justification of Cost: This meeting will provide me the opportunity to network with experts in the field of Implementation Science and learn from colleagues.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	185	185

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	1000	1500	2185	4685

Equipment

Type of cost: Other
VAT relief to be applied: True
Description: Laptop for PPM modelling and
Price excluding VAT £: 833
Expected life time: 36
Justification of Cost: Population Pharmacokinetic Modelling requires a computer with good processing power to undertake the modelling and simulation. Purchasing a laptop to undertake this, will make the modelling process efficient and the laptop will subsequently be used for the studies 2 and 3 as well.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	1000	0	0	1000

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	1000	0	0	1000

Consumables

Description: NONMEM software licence for Population Pharmacokinetic Modelling
Type of cost: Other
Please describe fully: NONMEM software licence for Population Pharmacokinetic Modelling
Justification of Cost: Population Pharmacokinetic Modelling requires the use of NONMEM - specialist software for study I. It has an annual license fee, which this cost will cover for study 1.

	Year 1	Year 2	Year 3	Total Cost (£)
	500	0	0	500

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	500	0	0	500

Patient and Public Involvement

Description: PPI support
Type of cost: Other
Please describe fully: PPI support
Justification of Cost: 2 days ad-hoc support and guidance. 4 members (£325/day).

	Year 1	Year 2	Year 3	Total Cost (£)
	867	867	867	2601

Description: 6 monthly PPIE panel meeting
Type of cost: Other

Please describe fully: 6 monthly PPIE panel meeting
Justification of Cost: 1 hour prep and 1 hour meeting (£50/meeting) - 7 in total.

	Year 1	Year 2	Year 3	Total Cost (£)
	467	467	467	1401

Description: PPI training to undertake role
Type of cost: Other
Please describe fully: PPI training to undertake role
Justification of Cost: This cost covers the fee for the Imperial PPI training. £35 per panel member. 4 in total. https://www.imperial.ac.uk/patient-experience-research-centre/ppi/ppi-training/

	Year 1	Year 2	Year 3	Total Cost (£)
	140	0	0	140

Description: PPI training time
Type of cost: Other
Please describe fully: PPI training time
Justification of Cost: Training needs assessment (4 hours) + 2 days to undertake PPI training to undertake role £325 per day

	Year 1	Year 2	Year 3	Total Cost (£)
	3000	0	0	3000

Description: Stakeholder meetings
Type of cost: Other
Please describe fully: Stakeholder meetings
Justification of Cost: 12 meetings - 2 hours (including preparation time). £50 / meeting

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1200	1200	2400

Description: PPI impact workshop
Type of cost: Other
Please describe fully: PPI impact workshop
Justification of Cost: In order to evaluate the impact of PPI in my fellowship, a cube framework exercise will be undertaken. 2 hour workshop - £50/person

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	200	200

Description: Refreshments for PPI impact workshop
Type of cost: Other
Please describe fully: Refreshments for PPI impact workshop
Justification of Cost: £15 per person (4 PPI members and researcher)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	75	75

Description: Co-produce patient / public facing materials
Type of cost: Other
Please describe fully: Co-produce patient / public facing materials
Justification of Cost: The PPI panel will co-develop and review materials for the website and Thrombosis UK. 2 days (£325/person)

	Year 1	Year 2	Year 3	Total Cost (£)
	867	867	867	2601

Description: Reviewing data, advising on take-home messages and reporting
Type of cost: Other
Please describe fully: Reviewing data, advising on take-home messages and reporting

Justification of Cost: During the course of my fellowship, my PPI panel will help with reviewing data, advise on take home messages and help with the final report writing. £325/person 2 days in total

	Year 1	Year 2	Year 3	Total Cost (£)
	867	867	867	2601

Description: Quality and Safety in Healthcare conference attendance to present findings from the work
Type of cost: Other
Please describe fully: Quality and Safety in Healthcare conference attendance to present findings from the work
Justification of Cost: The chair of the PPI panel will attend this conference with me to present the patient perspective on the findings from study 3. This cost covers the registration fee for their attendance.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1070	1070

Description: Travel to Quality and Safety in Healthcare conference for presentation with lead researcher
Type of cost: Other
Please describe fully: Travel to Quality and Safety in Healthcare conference for presentation with lead researcher
Justification of Cost: This covers the travel costs of the PPI member who attends the conference with me to present the patient perspective on the work completed.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	580	580

Description: Accommodation and food for Quality and Safety in Healthcare Conference
Type of cost: Other
Please describe fully: Accommodation and food for Quality and Safety in Healthcare Conference
Justification of Cost: This cost covers the accommodation and sustenance when attending the conference.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	350	350

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	6208	4268	6543	17019

Other Direct Costs

Description: Website for healthcare and patient group engagement
Type of cost: Other
Please describe fully: Website for healthcare and patient group engagement
Justification of Cost: A website will be established, which will allow patients, carers and members of the public, as well as healthcare professionals see what work is being done, how the UFH dosing protocol is being developed and what the outputs are. The website will have a patient / members of the public section and a healthcare professional section - with tailored information in each stream. The PPI panel in my research will co-produce the patient / members of the public section. This costs covers the establishment and maintenance of the website. I will be working with Kennedy Digital to do this task.

	Year 1	Year 2	Year 3	Total Cost (£)
	1701	1701	1701	5103

Description: E-gift cards for volunteers taking part in the Study 3 - phase I
Type of cost: Other
Please describe fully: E-gift cards for volunteers taking part in the Study 3 - phase I
Justification of Cost: 24 go-along interviews are to be completed as part of study 3 - phase I. This cost covers the e-gift cards which will be given to volunteers who take part (24 x £25).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	600	0	600

Description: E-gift cards for volunteers taking part in the Study 3 - phase II
Type of cost: Other
Please describe fully: E-gift cards for volunteers taking part in the Study 3 - phase II
Justification of Cost: During study 3 - phase II, 50 acceptability questionnaires will be administered at the pilot site following implementation of the contemporary UFH protocol. This cost covers the e-gift card costs for volunteers who take part and return a completed questionnaire (50 x £15).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	750	750

Description: E-gift cards for volunteers taking part in the Study 3 - phase III
Type of cost: Other
Please describe fully: E-gift cards for volunteers taking part in the Study 3 - phase III
Justification of Cost: As part of study 3 - phase III, the acceptability questionnaire will be re-administered at KCH (6 months post implementation) and at GSTT and RBHH (3 months post-implementation). This cost covers the e-gift cards which will be given to volunteers who take part and complete the questionnaire (150 (50 healthcare professionals from each hospital) x £15).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2250	2250

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	1701	2301	4701	8703

Training And Development

Type of cost: HEI
Description: Short Courses
Item Description: Implementation Science Course - University of Oxford
Justification of Cost: This cost covers the module fee for the implementation science module being undertaken at Oxford.

	Year 1	Year 2	Year 3	Total Cost (£)
	2890	0	0	2890

Type of cost: Other
Description: Overseas Research Visits
Item Description: Anticoagulation Forum - New Mexico - Practice development
Justification of Cost: As part of my practice development, I will be making a practice development trip to New Mexico to observe anticoagulation stewardship in action. Dr Alison Burnett, president of the Anticoagulation Forum will host me. This cost covers the travel, accommodation and food for the 5 day visit.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1872	0	1872

Type of cost: HEI
Description: Short Courses
Item Description: London School of Hygiene and Tropical Medicine (LSHTM) epidemiology modules
Justification of Cost: This cost covers the module registration fees for the 3 epidemiology modules being undertaken at the LSHTM, which will provide a solid foundation for the work being undertaken in study 2.

	Year 1	Year 2	Year 3	Total Cost (£)
	4207	2050	0	6257

Type of cost: Other
Description: Leadership Training Programme
Item Description: Coaching
Justification of Cost: As part of my personal effectiveness training I will undertake a series of coaching sessions in the final year of the my fellowship. This covers the costs of 10 x 2 hour coaching sessions.

	Year 1	Year 2	Year 3	Total Cost (£)

	0	0	1500	1500

Type of cost: Other
Description: Short Courses
Item Description: Travel to Oxford for the 5 day residential for the implementation science module
Justification of Cost: This cost covers the return train fare (return) from East Croydon - where I live to Oxford to undertake the 5 day residential for the implementation science module.

	Year 1	Year 2	Year 3	Total Cost (£)
	85	0	0	85

Type of cost: Other
Description: Short Courses
Item Description: Accommodation for 4 nights at Oxford for implementation science resdiential
Justification of Cost: This cost covers the 4 night stay at Oxford, on campus, to undertake the implementation science residential.

	Year 1	Year 2	Year 3	Total Cost (£)
	415	0	0	415

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	7597	3922	1500	13019

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	132246	126231	129169	387646

Indirect costs

HEI Indirect costs
No costs have been added

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	0	0	0	0

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	0	0	0
Total Higher Education Institution Costs (100%)s	9147	0	9147
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	35779	0	35779
Total NHS Research Costs	342720	0	342720
Total Research Costs	387646	0	387646

Requested From Funder			

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**114240**	**114240**	**114240**	**342720**
Travel-subsistence-and-conference-fees	**1000**	**1500**	**2185**	**4685**
Equipment	**1000**	**0**	**0**	**1000**
Consumables	**500**	**0**	**0**	**500**
Patient-and-Public-Involvement	**6208**	**4268**	**6543**	**17019**
Other-Direct-Costs	**1701**	**2301**	**4701**	**8703**
Training And Development	**7597**	**3922**	**1500**	**13019**
HEI-Indirect-costs	**0**	**0**	**0**	**0**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**132246**	**126231**	**129169**	**387646**

NHS Support Costs

No costs have been added

Have you discussed and agreed these costs with the Lead Network?	N/A

NHS Treatment Costs

No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient care being provided different from the usual standard treatment for the condition?	No

Usual Treatment Costs

No costs have been added**Excess Treatment costs/(savings)**

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

NHS Support and Excess Treatment Costs

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	387646
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	387646

11. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References - UFH protocol.pdf) is included as an appendix within this file.

(Figures / Tables - Figure 1 - Logic model - Unfractionated Heparin.pdf) is included as an appendix within this file.

(Research Timetable - Research timetable - UFH protocol.pdf) is included as an appendix within this file.

1

12. Participants and Signatories

The applicant is required to supply the names and email addresses (if not already registered on the ARAMIS application system) of the individuals who will be undertaking 'Participant' and 'Signatory' roles as part of the application. **Please note** that it is acceptable for the 'Head of Department' signatory to also act in another Participant role. If this is the case, they will need to be included in this section for both roles. Once the applicant has supplied their details, Participants and Signatories will be asked to confirm their participation, and in doing so acknowledge their involvement and input into this application and agree to be involved in it.

Participation must be confirmed by all Participants and Signatories before the application is submitted.

You must ensure that all Participants and Signatories are happy for your application to be submitted before submitting it on the online system.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Participants

Research Support

Research Support Participant 1

Title	Professor
Forename(s)	Bryony Dean
Surname	Franklin
Position	

Research Support Participant 2

Title	Professor
Forename(s)	Louise
Surname	Rose
Position	Professor in Critical Care Nursing

Research Support Participant 3

Title	Professor
Forename(s)	Joseph
Surname	Standing
Position	Professor of Pharmacometrics

Practice Support

Practice Support Participant 1

Title	Professor
Forename(s)	David
Surname	Taylor
Position	

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mrs
Forename(s)	Ann-Marie
Surname	Murtagh
Position	Director of R&I Operations

Signatories

Host Organisations Heads of Department

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Roopen
Surname	Arya
Position	Professor of Thrombosis and Haemostasis

Head of Department or Senior Manager 2

Title	Dr
Forename(s)	ben
Surname	forbes
Position	Reader

The ‘Head of Department’ signatories and Finance Participant can be working on their section of the application form whilst you (the applicant) works on the rest of the application.

Signatories must confirm their participation on your application and complete their input in the relevant sections of the application before you will be able to press the submit button.

Signatories must also approve your application after you select the ‘submit’ option but BEFORE the application submission deadline.

The signatory approval from your Heads of Department will result in the application being fully submitted to the

NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system generated email.

The NIHR will not accept any applications unless fully approved by your signatories prior to the deadline of 1:00pm on Wednesday 15th November 2023.

Please see the **'Application Submission Process Flow Diagram' (Annex A)** in the Guidance Notes for further information.

13. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
Not applicable.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the NIHR Research Governance Framework. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

14. Validation Summary

Dr ben forbes (Head of Department or Senior Manager)

Professor Roopen Arya (Head of Department or Senior Manager)

Please follow the next steps in order to complete your application submission process;

- **'Validate'** all mandatory/required fields listed below (that are required to be completed/amended before submitting)
- Click **'Save and Close'**
- Click the **'Submit'** option
- Application submission and all signatory approvals must be completed by **1:00pm on Wednesday 15th November 2023.**

You will receive an automated email containing an acknowledgement that we have received your application.

Application is valid

If there are no validation requirements above you may be ready to submit the application. To do so '**Save and Close**' the application and then select '**Submit**'.

Following your submission, the application will be available to your signatories for approval. **Signatories must approve your submitted application before the application deadline.**

Appendices

1. References - UFH protocol.pdf
2. Figure 1 - Logic model - Unfractionated Heparin.pdf
3. Research timetable - UFH protocol.pdf

Appendix #1: References - UFH protocol.pdf

References – UFH Protocol

1. World Health Organization Model List of Essential Medicines – 22nd List, 2021. Geneva: WHO
2. Cousins DH *et al*. Br J Clin Pharmacol 2012; 74(4): 597–604
3. U.S. Department of Health and Human Services, Office of Disease Prevention and Health Promotion 2014. National Action Plan for Adverse Drug Event Prevention. Washington, DC
4. Institute for Safe Medication Practices 2017. Medication Safety for High-Alert. Available at: https://www.ismp.org/assessments/high-alert-medications
5. Niccolai CS *et al*. Pharmacotherapy 2004; 24(8 Pt 2):146S–155S
6. Fanikos J *et al*. Am J Med 2023; 136(9): 927-936
7. Pino FA *et al*. Am J Health-Syst Pharm. 2019; 76:1972-1979
8. Elliott RA *et al*. BMJ Qual Saf 2021;30:96–105
9. Medication Without Harm - Global Patient Safety Challenge on Medication Safety. Geneva: World Health Organization, 2017
10. Hirsh J *et al*. Chest 2008; 133:141S–159S
11. Basu D *et al*. N Engl J Med 1972; 287: 324–327
12. Morabia A. Lancet 1986; 1: 1278–1279
13. Olson JD *et al*. Arch Path Lab Med 1998; 122: 782–798
14. Eikelboom JW & Hirsh J. Thromb Haemost 2006; 96: 547 – 552
15. Arachchillage DRJ *et al*. Thrombosis Research 2017; 157: 157–161
16. Guervil DJ *et al*. Ann Pharmaco 2011; 45: 861-868
17. Zaki HV *et al*. J Anesthes 2019; 11: 3
18. Coons JC *et al*. Am J Hematol. 2019; 94: 1015–1019
19. Swayngim R *et al*. Thromb Res 2021; 208: 18–25
20. Belk KW *et al*. JTH 2016; 14: 2148-2157
21. Raschke RA *et al*. Ann Int Med 1993; 119: 874–881
22. Whitman-Purves E *et al*. Clin App Thromb Hemost 2018; 24(2): 310-316
23. Williams-Norwood T *et al*. AACN Adv Crit Care. 2020; 31(2): 129–137
24. Kim JS *et al*. Clin App Thromb Hemost 2014; 20(7): 723-728
25. Kindelin NM *et al*. Ann Pharm 2021; 55(5): 575–583
26. McLaughlin K et al. Clin App Thromb Hemost 2019; 25: 1-6
27. Sheiner LB. Drug Metab Rev. 1984;15:153–171
28. Aarons L. Br J Clin Pharmacol. 1991;32:669–670
29. Speed V *et al*. JTH 2020; 18: 2296-2307
30. Wan M *et al*. Br J Clin Pharmacol 2022; 88: 1223–1234
31. Jia Z *et al*. J Trans Med 2015 13:45
32. Al-Sallami H *et al*. Br J Clin Pharmacol 2016; 82 178–184
33. Delavenne X *et al*. Br J Anaesth 2017; 118(5): 705–12
34. Schulman S *et al*. JTH 2022; 20(10): 2214-2225
35. Franklin BD *et al*. BMJ Qual Saf 2014; 23: 765–772
36. National Coordinating Council for Medication Error Reporting and Prevention 2001. Available at: http://www.nccmerp.org/types-medication-errors
37. Ishikawa K. Guide to Quality Control 1968, JUSE, Tokyo
38. Damschroder LJ *et al*. Impl Sci (2022) 17:75
39. Koldeweij C *et al*. PLoS ONE 2022; 17(5): e0267445
40. Carpiano RM. Health & Place 2009; 15: 263– 2
41. Kerr M. Res Nurs & Health 2009; 32: 647-656
42. Sekhon M *et al*. BMC Health Ser Res 2022; 22:279
43. Carroll C *et al*. Impl Sci 2007; 2: 40
44. Baglin T *et al*. Br J Haem 2006; 133: 19–34

Appendix #2: Figure 1 - Logic model - Unfractionated Heparin.pdf

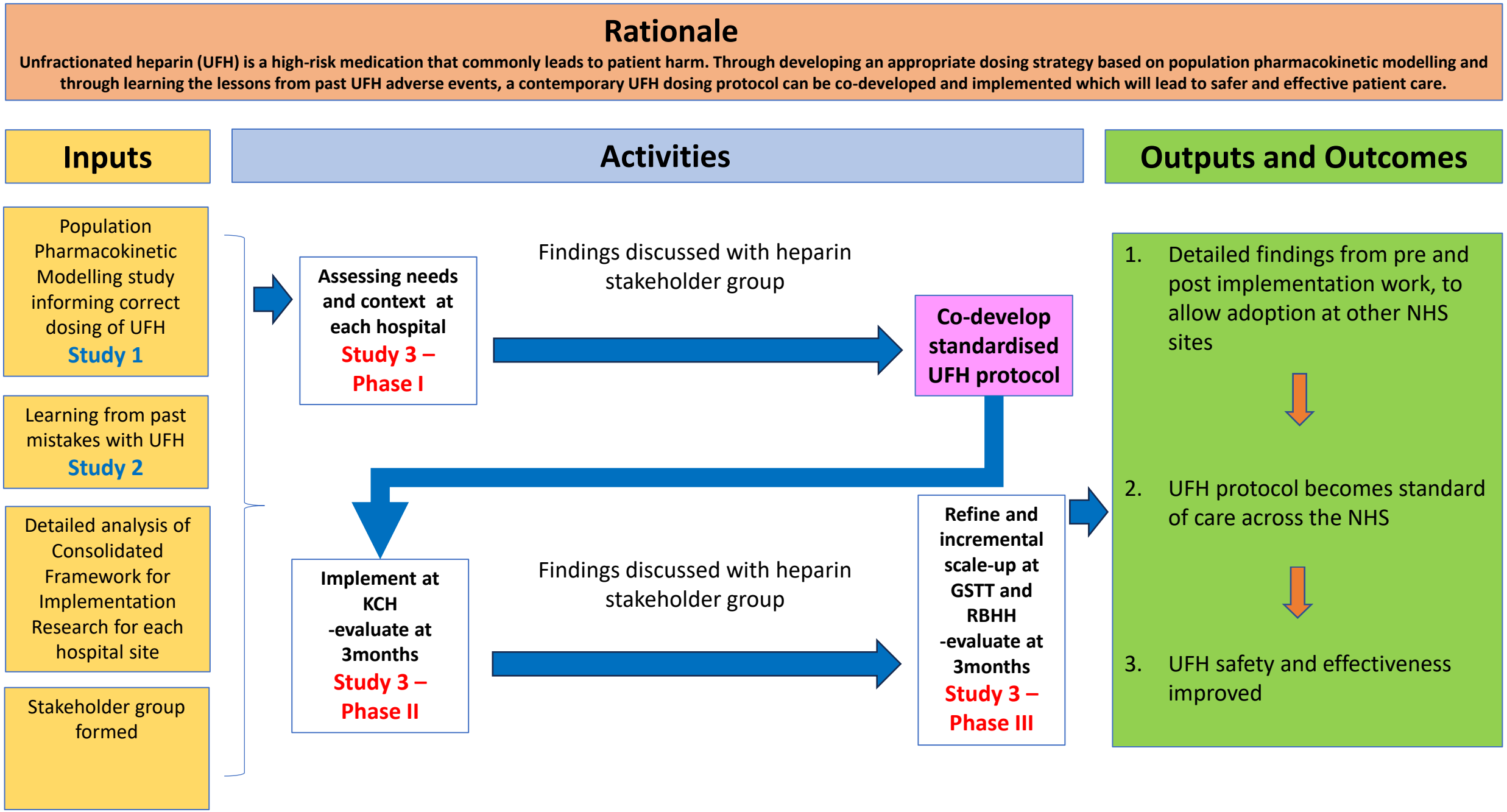
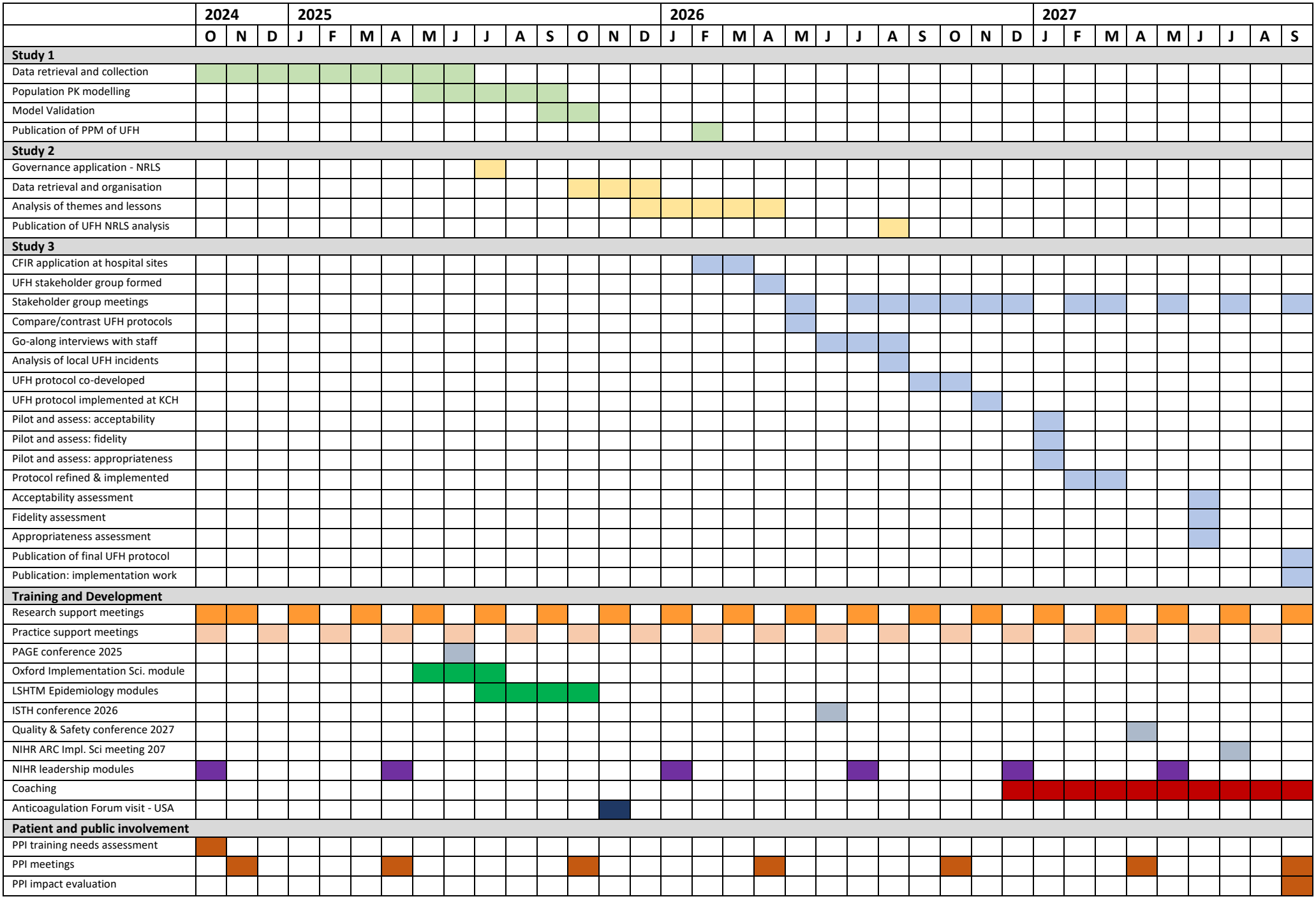


Figure 1 – Logic model for the implementation of the contemporary UFH infusion protocol at King's College Hospital (KCH), Guy's and St. Thomas' Hospital (GSTT) and Royal Brompton and Harefield Hospital (RBHH)

Appendix #3: Research timetable - UFH protocol.pdf


2024
2025
2026
2027
O N D J F M A M J J A S O N D J F M A M J J A S O N D J F M A M J J A S
Study 1
Data retrieval and collection
Population PK modelling
Model Validation
Publication of PPM of UFH
Study 2
Governance application - NRLS
Data retrieval and organisation
Analysis of themes and lessons
Publication of UFH NRLS analysis
Study 3
CFIR application at hospital sites
UFH stakeholder group formed
Stakeholder group meetings
Compare/contrast UFH protocols
Go-along interviews with staff
Analysis of local UFH incidents
UFH protocol co-developed
UFH protocol implemented at KCH
Pilot and assess: acceptability
Pilot and assess: fidelity
Pilot and assess: appropriateness
Protocol refined & implemented
Acceptability assessment
Fidelity assessment
Appropriateness assessment
Publication of final UFH protocol
Publication: implementation work
Training and Development
Research support meetings
Practice support meetings
PAGE conference 2025
Oxford Implementation Sci. module
LSHTM Epidemiology modules
ISTH conference 2026
Quality & Safety conference 2027
NIHR ARC Impl. Sci meeting 207
NIHR leadership modules
Coaching
Anticoagulation Forum visit - USA
Patient and public involvement
PPI training needs assessment
PPI meetings
PPI impact evaluation